



CENTRE FOR ADVANCED STUDIES
Dr. APJ Abdul Kalam Technical University, Lucknow

M. TECH - MECHATRONICS

Effective from 2022-23

Program Structure

The Centre for Advanced Studies (Dr. A.P.J. Abdul Kalam Technical University, Lucknow, U.P.) has started offering Master of Technology in Mechatronics from the academic session 2018-19. The curriculum has been developed considering the present and future needs of industry and higher education. The Centre will facilitate both industry ready and research based ambience to students with world class e-library, renowned faculty members to achieve academic excellence and other services.

This is a program aimed to provide the student with in-depth knowledge of domains in the field of Mechatronics.

M. Tech in Mechatronics. It is a multi-disciplinary study dealing with the integration of mechanical devices, actuators, sensors, electronics, intelligent controllers and computers. It is unique balance of rigorous fundamentals and engaging real-world applications in Mechatronics Engineering which provide the students the opportunity to design, build, operate and maintain machinery with mechanical, electronic/electrical and computer control systems.

The course structure will help students to develop knowledge and skill in the following proportionate:

90% Technical /Research	Industrial automation and robotics, Sensors and actuators, Control systems, Machine vision, Microcontrollers and embedded systems. Advanced microcontrollers, Mechatronics systems design, Machine tool control and condition monitoring, Flexible manufacturing system, Smart materials for mechatronics, Additive manufacturing and tooling, Vetrronics, Finite element method, Bio mechatronics, and Intelligent visual surveillance systems, IOT and smart manufacturing.
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	Excellent Visionary Skill that focus on scalability, cost effectiveness and implementation ease.
10% Business Process & Managerial Practices	Know the Business Dynamics, Business Processes and good planning, ability to work with all management level and resolve issues, Business need with automation Requirement. Consulting Skill, Communication Skill, Legal Understanding.

The students of Mechatronics program will attain following objectives:

- a) Students will understand the evolving Mechatronics systems from their underlying physical principles and properties.
- b) Students will design Mechatronics systems by applying underlying mathematical principles, supporting software for engineering model preparation and analysis.
- c) Students will be proficient in the systematic explorations of alternatives for Mechatronics systems design.
- d) Students will be proficient in the use of communications (oral presentations and written reports) to articulate their ideas effectively.
- e) Students will be prepared for the continuing learning and self-improvement necessary for a productive career in Mechatronics.

Program Structure:

M.Tech course is a full time two year program. The Program Structure is designed such that the students shall study core subjects and few elective subjects of Mechatronics Engineering. In Semester 1, students shall study fundamental core courses of Mechatronics Engineering while in the next two semesters students shall study in depth Advanced core and elective subjects. Semester IV is Thesis/ Dissertation.

Every core course consists of lecture (L) hours, tutorial (T) hours and practical (P) hours. Elective courses consist of Lecture (L) hours only. The credit I for a course is dependent on the number of hours of instruction per week in that course, as given below:

- (1) 1h/week of Lecture (L) = 1 credit
- (2) 2h/week of Practical (P) = 1 credit
- (3) 1h/week of Tutorial(T) = no credit
- (4) Credit for a Theory course = No. of hours of lectures per week = L
- (5) Credit for a Lab course = $\frac{1}{2}$ * No. of hours of lab per week = P

Course Credit Distribution

Category	Core Courses	Elective Courses	Labs	Project	Total Credit
Credits	21	12	5	22	60

SEMESTER	COURSE TYPES	CREDITS	
Semester I			
1	Core Course	(3 credit * 4 course) = 12	TOTAL COURSE = 5 TOTAL SEM CREDIT = 18
1	Elective Course	(3 credit * 1 course) = 3	
2	Core Course Lab and ISR-I	1 credit * 3 course = 3	
Semester II			
2	Core Courses	3 credit * 2 course = 6	TOTAL COURSE = 5 TOTAL SEM CREDIT = 17
2	Elective Course	3 credit * 3 course = 9	
2	Core Course Lab and ISR-II	1 credit * 2 course = 2	
Semester III			
3	Core Courses	3 credit * 1 course = 3	TOTAL COURSE = CREDIT = 12
3	Thesis/Dissertation- I	9 credit	
Semester IV			
4	Thesis/ Dissertation-II	13 credit	TOTAL COURSE = 1 MAJOR PROJECT TOTAL SEM CREDIT = 13
TOTAL PROGRAM CREDIT			60

MECHATRONICS PROGRAM

VISION

To recognize itself into a world class Centre of excellence in the field of Mechatronics for Postgraduate Degree as well as cutting edge Research and Development for the global good.

MISSION

- M1.** To impart quality education for Postgraduate Studies meeting the needs of industry and society, and achieve excellence through research and development.
- M2.** To provide a stimulating platform for creating and disseminating research based knowledge/technologies through innovation and entrepreneurship for the development of State/Country.
- M3.** To facilitate quality research work in the field of Mechatronics by practicing professional ethics and team-spirit.
- M4.** To promote interactions of the department with alumni, institutions, industries and other stake-holders through networking and collaboration.

PEOs

- PEO1.** Graduates will possess theoretical knowledge as well as professional skills in the areas of Mechatronics, to pursue a career in leading organizations.
- PEO2.** Graduates will demonstrate active involvement and participation in Research and Development leading to innovative contributions at both national and international levels.
- PEO3.** Graduates will engage in lifelong learning with a strong inclination towards research and development by pursuing higher studies as a competent professional.
- PEO4.** Graduates will work with values and professional ethics to create and disseminate optimized solutions for addressing the challenges faced by industry/society.

PROGRAM OUTCOMES

1. **PO1:** An ability to independently carry out research /investigation and development work to solve practical problems
2. **PO2:** An ability to write and present a substantial technical report/document.
3. **PO3:** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
4. **PSO 1:** An ability to apply advanced knowledge in the area of Mechatronics, to design and implement efficient, effective and optimized solutions targeting the real problems of the industry through research and development.
5. **PSO 2:** An ability to develop innovative, sustainable and adaptive solutions for catering the needs of the society by applying standard principles in computing and professional ethics.

SEMESTER –I

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	AMTR-101	Advanced Engineering Mathematics	3	0	0	3
2	AMTR-102	Sensors and Actuators	3	0	0	3
3	AMTR-103	Control System: Modeling, Design and Analysis	3	0	0	3
4	AMTR-104	Microcontrollers and Embedded Systems	3	0	0	3
5	AMTR-105/ AMTR-106	Concepts of Mechanical Engineering/ Concepts of Electronics and Electrical Engineering	3	0	0	3
6	AMTR-151	Lab-I Sensors, Drives and Control lab	0	0	2	1
7	AMTR-152	Lab-II Microcontroller and Embedded systems lab	0	0	2	1
8	AMTR-153	Independent Study & Research-I (ISR-I)	0	0	2	1
		TOTAL	15	0	6	18

SEMESTER –II

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	AMTR-201	Industrial Automation	3	0	0	3
2	AMTR-202	Industrial Robotics	3	0	0	3
3		Elective-I	3	0	0	3
4		Elective- II	3	0	0	3
5		Elective- III	3	0	0	3
6	AMTR-251	Lab-III (Robotics and Automation lab)	0	0	3	1

7	AMTR-252	Independent Study & Research-II (ISR-II)	0	0	3	1
		TOTAL	15	0	6	17

SEMESTER –III

S N	Course Code	Subject	Periods (Hrs/week)			Credits
1	AMTR-301	Research Methodology and Technical Writing	3	0	0	3
2	AMTR-351	Thesis/Dissertation- I	0	0	18	9
3		Teaching/Lab Assignments & evaluation	0	0	4	0
		TOTAL	3	0	22	12

SEMESTER –IV

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	AMTR-451	Thesis / Dissertation - II	0	0	30	13
		TOTAL	0	0	30	13

Elective Subjects-

Elective- (I-III)

S. No.	Course code	Subject
1	AMTR-211	Mechatronics Systems Design
2	AMTR-212	PC based automation

3	AMTR-213	Signal Processing in Mechatronics Systems
4	AMTR-214	Control of Mechatronic Systems
5	AMTR-215	Machine tool control and condition monitoring
6	AMTR-216	Mechatronics in Metrology
7	AMTR-217	Machine vision
8	AMTR-218	Design of Mechanisms and machines
9	AMTR-219	Artificial Intelligence & Expert System in Automation
10	AMTR-220	Modeling and Simulation of Mechatronic Systems
11	AMTR-221	Advanced microcontroller and embedded systems
12	AMTR-222	Finite element method
13	AMTR-223	Mechatronics in CNC
14	AMTR-224	Theory of Machining
15	AMTR-225	Computer Aided Manufacturing
16	AMTR-226	Flexible Manufacturing System
17	AMTR-227	Rapid Prototyping
18	AMTR-228	MEMS and NEMS
19	AMTR-229	Smart materials for mechatronics applications
20	AMTR-230	Optimization Techniques & Design of Experiments
21	AMTR-231	Additive Manufacturing & Tooling
22	AMTR-232	Vetronics
23	AMTR-233	Computational fluid dynamics

24	AMTR-234	Intelligent visual surveillance systems
25	AMTR-235	Bio-Mechatronics
26	AMTR-236	Internet of Things and Smart Manufacturing
27	AMTR-237	Augmented Reality
28	AMTR-238	Drone Technology

Labs

S No.	Lab code	Lab Name
1	AMTR-151	Sensors, Drives and Control lab
2	AMTR-152	Microcontroller and Embedded systems lab
3	AMTR-153	Independent Study & Research-I (ISR-I)
4	AMTR-251	Robotics and Automation lab
5	AMTR-252	Independent Study & Research-II (ISR-II)
6	AMTR-351	Thesis/Dissertation- I
7	AMTR-451	Thesis / Dissertation - II



CENTRE FOR ADVANCED STUDIES

Dr. APJ Abdul Kalam Technical University, Lucknow

SYLLABUS FOR M.TECH (MECHATRONICS) Effective from 2022-23

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S. No.	Title	Page No.
1	Semester- I	12
2	Semester- II	20
3	Semester- III	58
4	Semester- IV	60

MARKS DISTRIBUTION

Following is marks distribution for courses depending upon the credit of the courses:

(1 credit = 50 marks)

Credit	Type of course	Max. Marks	Major (End semester examination)	Performance evaluation		Lab evaluation
				Minor I & II	Internal Assessments (Attendance, Quiz, Seminar & Assignments)	
1	Lab course	50	25	25		-
2	Theory course	100	50	30 (15 each)	20	-
3	Theory course	150	75	40 (20 each)	35	-
4	Theory course with lab	200	75	40 (20 each)	35	50
4	Thesis / dissertation-I	450	300	150		-
5	Thesis / dissertation-II	650	400	250		-

Note: Maximum Marks are fixed for all the courses. Only internal assessments and lab evaluation marks distribution may vary as per the instructions communicated by the competent authority. Refer to M.Tech Ordinance & Regulation for Assessment Procedures and Performance Evaluation.

SEMESTER-I

CORE COURSES

AMTR-101	ADVANCED ENGINEERING MATHEMATICS	L	T	P	C
		3	0	0	3
Prerequisite	NIL				

Syllabus:

Linear Algebra: Matrix algebra; basis, dimension and fundamental subspaces; solvability of $Ax = b$ by direct Methods; orthogonality and QR transformation; eigenvalues and eigenvectors, similarity transformation, singular value decomposition, Fourier series, Fourier Transformation, FFT.

Vector Algebra & Calculus: Basic vector algebra; curves; grad, div, curl; line, surface and volume integral, Green's theorem, Stokes's theorem, Gauss-divergence theorem.

Differential Equations: ODE: homogeneous and non-homogeneous equations, Wronskian, Laplace transform, series solutions, Frobenius method, Sturm-Liouville problems, Bessel and Legendre equations, integral transformations; PDE: separation of variables and solution by Fourier Series and Transformations, PDE with variable coefficient.

Numerical Technique: Numerical integration and differentiation; Methods for solution of Initial Value

Problems, finite difference methods for ODE and PDE; iterative methods: Jacobi, Gauss-Siedel, and successive over-relaxation.

Complex Number Theory: Analytic function; Cauchy's integral theorem; residue integral method, conformal mapping. Statistical Methods: Descriptive statistics and data analysis, correlation and regression, probability distribution, analysis of variance, testing of hypothesis.

Optimization Techniques: Calculus of several variables, Implicit function theorem, Nature of singular points, Necessary and sufficient conditions for optimization, Constrained Optimization, Lagrange multipliers, Gradient method – steepest descent method

References and Text Books:

1. H. Kreyszig, "Advanced Engineering Mathematics", Wiley, (2006).
2. Gilbert Strang, "Linear Algebra and Its Applications", 4th edition, Thomson Brooks/Cole, India (2006).
3. J. W. Brown and R. V. Churchill, "Complex Variables and Applications", McGraw-Hill Companies, Inc., New York (2004).
4. J. W. Brown and R. V. Churchill, "Fourier Series and Boundary Value Problems", McGraw-Hill Companies, Inc., New York (2009).
5. G. F. Simmons, "Differential Equations with Applications and Historical Notes", Tata McGraw-Hill Edition, India (2003).
6. S. L. Ross, "Differential Equations" 3rd edition, John Wiley & Sons, Inc., India (2004).
7. K. S. Rao, "Introduction to Partial Differential Equations", PHI Learning Pvt. Ltd (2005).
8. R. Courant and F. John, "Introduction to Calculus and Analysis, Volume I and II", Springer-Verlag, New York, Inc. (1989).
9. K. Atkinson and W. Han, "Elementary Numerical Analysis" 3rd edition, John Wiley & Sons,

Inc., India (2004).

10. R. A. Johnson and G. K. Bhattacharya, “Statistics, Principles and Methods”, Wiley (2008).

Course Objectives

At the end of this course, students should:

1. Introduce students to ordinary differential equations and the methods for solving these equations, using differential equations as models for real-world phenomena, and Integrate the knowledge accumulated in the calculus to solve applied problems
2. To apply the mathematical methods to solve the engineering problem
3. Provide a rigorous introduction to upper-level mathematics which is necessary for students of engineering, physical sciences, and mathematics.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-101.1	Identify the powerful mathematical approaches to solve complex engineering problems.
AMTR-101.2	Apply the appropriate mathematical theories and computational tools to solve mathematical problems.
AMTR-101.3	Students can remember terminologies and formulae in matrices, complex numbers, differential calculus, vector Algebra & calculus, differential Equations, Numerical techniques, and complex number theory.
AMTR-101.4	Discuss the appropriate application of mathematical theory in the solution of engineering problems.
AMTR-101.5	Students can find the matrices to solve systems of equations, work with and solve homogeneous and non-homogeneous ODEs and systems of ODEs and recognize ODEs of varying order and use these to solve problems involving population dynamics, oscillation of a spring and resistance in a circuit, etc.

AMTR-102	SENSORS AND ACTUATORS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Brief overview of measurement systems, classification, characteristics and calibration of different sensors.

Measurement of displacement, position, motion, force, torque, strain gauge, pressure flow, temperature sensor sensors, smart sensor. Optical encoder, tactile and proximity, ultrasonic transducers, opto-electrical sensor, gyroscope. Principles and structures of modern micro sensors, micro-fabrication technologies: bulk micromachining, surface micromachining, LIGA, assembly and packaging.

Pneumatic and hydraulic systems: actuators, definition, example, types, selection. Pneumatic actuator. Electro-pneumatic actuator. Hydraulic actuator, control valves, valve sizing valve selection. Electrical actuating systems: solid-state switches, solenoids, voice coil; electric motors; DC motors, AC motors, single phase motor; 3-phase motor; induction motor; synchronous motor; stepper motors. Piezoelectric actuator: characterization, operation, and fabrication; shape memory alloys.

References and Text Books:

1. John G. Webster, Editor-in-chief, "Measurement, Instrumentation, and Sensors Handbook", CRC Press (1999).
2. Jacob Fraden, "Handbook of modern Sensors", AIP Press, Woodbury (1997).
3. Nadim Maluf, "An Introduction to Microelectromechanical Systems Engineering", Artech House Publishers, Boston (2000).
4. Marc Madou, "Fundamentals of Microfabrication", CRC Press, Boca Raton (1997).
5. Gregory Kovacs, "Micromachined Transducers Sourcebook", McGraw-Hill, New York (1998).
6. E. O. Deobelin and D. Manik, "Measurement Systems – Application and Design", Tata McGraw-Hill (2004).
7. D. Patranabis, "Principles of Industrial Instrumentation", Tata McGraw-Hill, eleventh reprint (2004).
8. B. G. Liptak, "Instrument Engineers' Handbook: Process Measurement and Analysis", CRC (2003).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to understand the classification of different types of measurement systems their static and dynamic characteristics.
2. Different types of sensors and actuators with their working principles for different domain of applications.
3. To provide skills to design a mechatronics system with the integration of sensors and actuators.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-102.1	Understand the variety of measurement systems and their characteristics. Study of some basic sensors.
AMTR-102.2	Study the working principle and development of advanced sensors based on the different applied principles.
AMTR-102.3	Different micro-fabrication technologies for design and development of a sensors and other MEMS devises.
AMTR-102.4	Study of different mechanical actuating system.
AMTR-102.5	Study of different electrical actuating system. Characterization, operation, and fabrication of piezoelectric actuator.

AMTR-103	CONTROL SYSTEM: MODELING, DESIGN AND ANALYSIS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction and need for Control Systems, Block Diagram reduction, Signal flow graph, Mason's gain formula, Mathematical modelling of electrical, mechanical, and electro-mechanical systems.

Laplace transforms, transfer functions, electrical analogues of other dynamical systems. State-space modelling of dynamical systems, Time response of First & Second order systems, Time domain specifications - steady state errors and error constants.

Stability, Definition of stability, Routh-Hurwitz test, Lyapunov theory, Properties of feedback, Basic idea of feedback control systems, Error analysis, Stability analysis using Bode Plot, Polar Plot, Nyquist stability criterion, gain and phase margins, robustness.

Design and implementation of on-off, P, PI, PD, and PID controllers, Modified PID Controller, Tuning of controllers, The root-locus technique, Design of controllers using root-locus, Pole placement with state feedback, controllability.

Design and analysis of Lead, Lag, Lead-Lag/Lag-Lead compensators, Control system design using Frequency domain analysis, Servo motor-mathematical modeling, analysis and implementation of controllers for servo motor.

References and Text Books:

1. K. Ogata, Modern controls engineering "Prentice Hall of India Pvt. Ltd., New Delhi, 2005.
2. B.C. kuo, "Automatic Control Systems", Prentice Hall of India Pvt. Ltd., New Delhi, 2004
3. I.J. Nagrath and Gopal. "Control system engineering", new age international (P) Ltd., 2006.
4. M. Nakamura. S. Gata & N. Kyura, Mechatronic servo system control. Springer 2009.

Course Objectives

At the end of this course, students should:

1. To introduce different types of system and identify a set of algebraic equations to represent and model a complicated system into a more simplified form to interpret different electrical and mechanical systems to construct models for analysis.
2. To employ time domain analysis to predict and diagnose transient performance parameters of the system for standard input functions and identify the needs of different types of controllers and compensator to ascertain the required dynamic response from the system.
3. Formulate different types of analysis in frequency domain to explain the nature of stability of the system.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-103.1	Understand the need and basics of control system with the help of block diagrams and mathematical model formation of a given plan.
AMTR-103.2	Apply appropriate analytical techniques to calculate the time domain specifications under steady state and transient conditions.
AMTR-103.3	Time and frequency domain Analysis of Lead, Lag, and Lag-Lead compensators to design the desired control system.
AMTR-103.4	Analyse the frequency response plots such as Bode plot and Polar plots to develop the controller for a specific process control system.
AMTR-103.5	Assess current research issues and future directions in field of servo motor control and process control systems .

AMTR-104	MICROCONTROLLERS AND EMBEDDED SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction to Embedded Systems and microcomputers: Introduction to Embedded Systems, Embedded System Applications, Block diagram of embedded systems, Challenges for Embedded system Design, Evolution of computing systems and applications. Classification of the processors used in embedded design, Basic Computer architecture: Von- Neumann and Harvard Architecture. Basics Processing Cores – RISC & CISC. Computing performance, Throughput and Latency, Microcomputer applications to embedded systems and Mechatronics.

Microcontroller:

Introduction to Microcontroller and its families, Criteria for Choosing Microcontroller. Microcontroller Architecture, Programming model, Addressing modes, Instruction sets, Assembly and C programming for Microcontroller, I/O programming using assembly and C language, Interrupt Controller, I/O interfacing, Timers, Serial Communication, LCD Controller. Different types of Sensors.

Microcontroller Interfacing: Introduction to Microcontroller Interfacing and applications: case studies: Display Devices, Data Acquisition Systems, controllers and Drivers for DC, Servo and Stepper Motor.

Introduction to MSP 430: Memory Model, Overview of Instructions, I/O Ports, Timers and Interrupts.

Introduction to Advanced Embedded Processor and Software: ARM Processor, Real Time Operating System (RTOS), Embedded C.

References and Text Books:

1. Introduction to Embedded Systems: Shibu K V, McGRAW Hill Publications.
2. Embedded Systems: Raj Kamal, TATA McGRAW Hill Publications
3. Computer System Architecture: M. Morris Mano.
4. PIC Microcontrollers and Embedded Systems: M. A. Mazidi, R.D. Mckinlay and D. Casey, Pearson Publications
5. Programming and Customizing the PIC Microcontroller: M. Predko, McGRAW Hill Publications.
6. Embedded C Programming and Microchip PIC: R. Barnett, L. O’Cull and S. Cox.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in embedded system design. (Apply)
2. Analyze the microcontroller programming model, data acquisition systems and motor drivers. (Analysis)
3. To provide skills to interface motors, display devices and peripherals to the ARM processor. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-104.1	Understand the embedded system design and configuration. To study various microcontroller applications in mechatronic systems.
AMTR-104.2	Apply appropriate techniques to choose suitable sensors and display for microcontroller based process control system. Develop programming skills.
AMTR-104.3	Analyse different interfacing techniques for display devices and motors.
AMTR-104.4	Design microcontroller based prototype systems for specific control problems.
AMTR-104.5	Assess current research issues and future directions in embedded system design.

AMTR-105	CONCEPTS OF MECHANICAL ENGINEERING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Mechanisms: Definition – Machine and Structure – Kinematic link, pair and chain – classification of Kinematic pairs – Constraint & motion - Degrees of freedom, Slider crank mechanisms.

Power Screws: Forms of threads, multiple threads, Efficiency of square threads, Design of screw jack.

Friction drives: Introduction, belt and rope drives, open and crossed belt drives, power transmission.

Gearing and cams: Gear profile and geometry-nomenclature of spur and helical gears, law of gearing simple and compound gear trains.

Fluid mechanics: Fluid and continuum, Physical properties of fluids, Rheology of fluids, Pressure-density-height relationship, manometers. Types of fluid flows, Bernoulli's equation and its applications.

Strength of materials: Introduction, normal stress and strain, shear stress and strain, stresses on inclined sections, strain energy, impact loads and stresses, state of plane stress, principal stress and strain, maximum shear stress.

Thermodynamics: Fundamental laws, Temperature, Enthalpy, Entropy. Internal combustion engines.

References and Text Books:

1. Joseph Edward Shigley, Charles R. Mischke, "Mechanical Engineering Design" McGraw Hill International Edition, 2008.
2. Juvinall, R.C., Fundamentals of Machine Component Design, John Wiley and Sons, 1994.
3. Spottes, M.F., Design of Machine elements, Prentice-Hall India, 1994.
4. Deutschman, D., Michels, E., Machine Design Theory and Practice, Macmillan, 1992
5. Collens, J.A., "Failure of Materials in Mechanical Design", John Wiley & Sons, 1991.
6. Fuchs H.O., "Metal fatigue in Engineering", John Wiley & Sons, 1991.
7. Thermodynamics, P.K. Nag McGraw Hill Education.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of different mechanisms for mechanical machines. (Apply)
2. Analyze how power screws, friction drives and gears is used to transmit power in machines. (Analysis)
3. To provide skills to design machines using knowledge of fluid mechanics, strength of materials and thermodynamics. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-105.1	Understand different types of mechanism and need in various application.
AMTR-105.2	Apply knowledge of power screws, friction drives and gears in power transmission.
AMTR-105.3	Analyse fluid properties in different conditions and calculation of those properties.
AMTR-105.4	Design advanced mechanisms based machines for greater efficiency and effectiveness.
AMTR-105.5	Assess current issues and future challenges in the field of motion and power transmission.

AMTR-106	CONCEPTS OF ELECTRONICS AND ELECTRICAL ENGINEERING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Digital- number system, logic gates and their applications, K-map, Combinational Logic circuits, Synchronous and Asynchronous Sequential logic circuits, Registers and counters, Memory and programmable logic.

Oscillators, Brief introduction of Transformer, Solid state devices- SCR, IGBT, Converters & Invertors, Variable speed drives (AC & DC). Test and measuring instruments-Measurement of voltage, current, frequency and power using multi meters, oscilloscopes, recorders, data loggers, signal sources, analysers and printers.

Microprocessor (8085/8086) architecture and its operations, pins description, programming model, basic interfacing concepts. Memory Interfacing, Programmable Peripheral Interfacing, Programmable I/O, Programmable Controllers, Programmable Timers, Data acquisition Interfacing: ADC, DAC, Serial and parallel data Communication interfacing.

References and Text Books:

1. Millman Halkias, Integrated Electronics, Tata McGraw-Hill
2. M. Morris Mano, Digital Design, McGraw-Hill
3. Gayakwad, Operational Amplifier-Linear Integrated Circuits, Prentice Hall India
4. Mohammad H. Rasid, Power Electronics, Prentice Hall India
5. Helfrick A.D and Cooper .W. D., Modern Electronic Instrumentation and Measurements Techniques, Prentice Hall 2008.
6. Leach, Digital Principles and Applications, ed. 7, TMH 2008.
7. Mandal, Digital Electronics: Principles and Applications, TMH 2009.
8. 8086 Microprocessors and Interfacings: D. Hall, TATA McGRAW Hill.
9. The Intel Microprocessors: B. Brey, Prentice Hall Publications.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in digital system design. (Apply)
2. Analyze how digital systems can be used as an effective technique in solving real world problems. (Analysis)
3. To provide skills to design basic electronic circuits such as measuring instruments, power converters, ADC, DAC etc. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-106.1	Understand the philosophy of the digital signal, positive and negative logic, Boolean algebra, logic gates, and number systems.
AMTR-106.2	Apply the fundamental scientific principles to understand the operation of transformers, switching devices and drivers.
AMTR-106.3	Analyse the measurement results obtained from different measuring instruments such as ammeter, voltmeter, multi-meter and oscilloscope.
AMTR-106.4	Design microcontroller based systems to solve the specific problems.
AMTR-106.5	Assess current research issues and future directions in field of data communication interfacing.

S. No. 1: (LAB-I)

AMTR-151	LAB-I SENSORS, DRIVES AND CONTROL LAB	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Experiments:

1. Introduction about Sensor and its basic working principles.
2. Introduction to basic network theorems & circuits:
 - Norton theorem
 - Thevenin's theorem
 - Wheat stone bridge
 - KLC, KVL, and Maximum Power Transfer Theorem.
3. Introduction of Position, Proximity, Distance & Velocity Measuring Sensors. (Ultrasonic Sensor, IR Sensor, Radar Sensor, LVDT etc.)
4. Introduction of Different Touch and Pressure type Sensor.
5. Introduction of Force and Torque Measuring Sensor.
6. Introduction of Temperature Sensor. (To Measure the Characteristics of RTD, Thermocouple and LM35 Sensor)
7. Introduction of Flow Measuring Sensor.
8. Introduction of different rays, Waves and chemical Detecting Sensors. (Color, LDR, solar, etc.)
9. Introduction to different Actuators (Electrical, Pneumatic, Hydraulic). (SMA, DEA, Piezoelectric, motors)
10. DIY (Do It Yourself) project on sensors and actuators model/prototype design/development; using these develop a system contain least two sensors and two actuators.

Text Books:

- [1].John G. Webster, Editor-in-chief, "Measurement, Instrumentation, and Sensors Handbook", CRC Press (1999).
- [2].Jacob Fraden, "Handbook of modern Sensors", AIP Press, Woodbury (1997).
- [3].Nadim Maluf, "An Introduction to Microelectromechanical Systems Engineering", Artech House Publishers, Boston (2000).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge for understanding the operation of various sensors used in automation industry.
2. Develop the ability to analyse, and debug the electrical circuits using fundamental circuit theory.
3. Develop the skill to design circuits for interfacing the sensors to solve real world problems.

Course Outcome:

Upon completion of their academic work student will be able:

AMTR -151.1	To understand the basic working principle of sensors and drives systems.
AMTR -151.2	To understand the basic network theorems and analyse the drive and sensor interfacing circuits.
AMTR -151.3	To understand the working principle of different touch and pressure type sensors, and the design of sensor interfacing circuits.
AMTR -151.4	Analyse how Electrical, Pneumatic, and Hydraulic actuators operate and develop the ability to debug various sensors and actuator circuits.
AMTR -151.5	Develop the prototype of real world automation project using sensors and actuators.

AMTR-152	LAB-II MICROCONTROLLER AND EMBEDDED SYSTEMS LAB	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Experiments:

1. Identify Mechatronics system and exploring different components using internet browsing.
2. Understanding different measuring device (DSO, Function Generator, Millimeter, and other measuring devices).
3. Understanding different basic electronics components (semiconductor, diode, transistor, resistor, op-amp, Mosfet, relay, 555 timer, motor driver).
4. Understanding different basic electrical components. (circuit and electrical components (resistance, inductor, capacitor), transformer, motors, generator)
5. Understanding different basic mechanical components. (gear, degree of freedom, mechanism, cam & follower, valves, flywheel)
6. Introduction about different software like- Solid work, Ansys, Mat lab, Proteus, Arduino IDE, etc.
7. Application and implementation of different basic electronics components to design circuit and mechatronics system. (Different logic Gate using transistor, Rectifier, amplifier, timer, etc.)
8. Introduction about different microcontroller and microcontroller board.(8051, arduino, xbee, Node ESU, raspberry pi, FPGA)
9. Developing Mini project.(present idea, design, implement , test, report)

Text Books

- [1].Jonathan W. Valvano, ‘Embedded Systems: Introduction to ARM Cortex-M Microcontrollers’, 2016.
- [2].Mohammed, Moaml, and Alves, Claudia, “Arduino Reference: Syntax, Concepts, and Examples - 2nd Edition” LLC - KDP Print US, 2021.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in programming to understand the microcontroller programming and design simple embedded systems.
2. Develop the ability to analyse, debug and verify the embedded programs using a simulator and on the real microcontroller.
3. Develop the problem solving skill to solve real world problems using Finite State Machine.

Course Outcome:

Upon completion of their academic work student will be able:

AMTR-152.1	To understand the basic concepts of mechatronics systems and explore different components of mechatronics systems.
AMTR -152.2	To understand different measuring devices such as DSO, Function Generator, Multi-meter, and basic electronic components such as semiconductor, diode, transistor, op-amp, relay, timer and motor drivers.
AMTR -152.3	To understand the operation of different basic mechanical components such as gear, valves, flywheel etc.
AMTR -152.4	To develop the skill to effectively use different software like- Solid work, Ansys, Matlab, Proteus, Arduino IDE, etc.
AMTR -152.5	To develop the ability to use different microcontroller and development boards (8051, arduino, xbee, Node ESU, raspbary pi, FPGA) to implement automation projects.

AMTR-153	Independent Study & Research-I (ISR-I)	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Activities:

In this course, student will do a variety of activities.

1. Student will learn how to read technical papers and how to analyze them.
2. Students will identify a small problem to work with after literature survey.
3. Students will prepare a detailed report of the works done by him.
4. Progress will be checked once a month for support and direction. A mentor will be assigned to each student for supervision.

Course Objective

The main objectives of course are:

1. The students will learn how to identify problems for the projects and how to start research work.
2. The student will learn reading technical papers and to do literature survey.
3. The purpose of this course is to create research aptitude in the students.

Course Outcome

Upon completion of their academic work student will be able:

AMTR -153.1	Understand basic concepts of literature review.
AMTR -153.2	Apply literature review in identification of a research problem.
AMTR -153.3	Analyze the various technical papers related to particular research problem.
AMTR -153.4	Design a system for better understanding and fast analysis of the technical papers.
AMTR-153.5	Assess challenges in identification of research problems through literature survey.

SEMESTER- II
CORE COURSES

AMTR-201	INDUSTRIAL AUTOMATION	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction: Automation in Production System, Principles and Strategies of Automation, Basic Elements of an Automated System, Advanced Automation Functions, Levels of Automations. Production Economics.

Material Transport: Automated Flow lines, Methods of work part Transport, Transfer Mechanism, and Buffer Storage.

Material handling and Identification Technologies: The material handling function, Types of Material Handling Equipment, Analysis for Material Handling Systems, Design of the System, Conveyor Systems, Automated Guided Vehicle Systems. Automated Storage Systems: Storage System Performance, Automated Storage/Retrieval Systems, Work-in-process Storage, Product identification system: Barcode, RFID etc.

Automated Assembly Systems: Design & Types of Automated Assembly Systems. Different control technologies in automation:

Automated Inspection and Testing: Automated Inspection Principles and Methods, Sensor Technologies for Automated Inspection, Coordinate Measuring Machines, Other Contact Inspection Methods, Machine Vision and other optical Inspection Methods.

References and Text Books:

1. “Automation, Production Systems and Computer Integrated Manufacturing”- M. P. Grover, Pearson Education.
2. “Computer Based Industrial Control” – Krishna Kant, EEE-PHI
3. Principles and Applications of PLC – Webb John, Mcmillan 1992
4. “An Introduction to Automated Process Planning Systems” – Tiess Chiu Chang & Richard A. Wysk
5. “Anatomy of Automation” – Amber G.H & P.S. Amber, Prentice Hall.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of automation in industrial manufacturing systems. (Apply)
2. Analyze how automation can be used as an effective tool for increasing efficiency and productivity of the industries. (Analysis)
3. To provide skills to design advanced automation based manufacturing systems. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-201.1	Understand basic concepts of automation, levels of automation, use of automation in production system and production economics.
AMTR-201.2	Apply appropriate automation techniques to solve industry problems.
AMTR-201.3	Analyze different materials handling and product identification techniques in automation.
AMTR-201.4	Design automated assembly systems, inspection and testing methods to solve the specific need of automation in industries.
AMTR-201.5	Assess current s and future challenges in the field of industries automation.

AMTR-202	INDUSTRIAL ROBOTICS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Fundamentals of robotics: Introduction, classification of robotic systems and applications.

The robot and its peripherals: Control systems and components, robot motion analysis and control, end effectors, feedback systems, encoders kinematics, homogeneous coordinates solution of the inverse kinematic problem, multiple solutions, Jacobian, work envelopes. Trajectory planning, Joint Interpolated Trajectory, Link joints and their Manipulator dynamics and force control.

Sensors in Robotics: Developments in sensor technology, sensory control Vision, ranging, laser, acoustic, tactile.

Programming Language: Industrial robot programming languages. Mobile robots, robot avoiding system, walking devices. Robot programming environment.

Robot applications: Application of robots in surgery, Manufacturing industries, space and underwater.

Humanoid robots, Micro robots, Social issues and Future of robotics.

References and Text Books:

1. K.S Fu, R.C. Gonzalez, C.S.G. Lee, Robotics, McGraw Hill, 1987.
2. Y. Koren, Robotics for Engineers, McGraw Hill, 1985.
3. J.J. Craig, Robotics, Addison-Wesley, 1986.
4. Saeed B. Niku, "Introduction to Robotics – Analysis, Systems and Application": PHI 2006.
5. Richard D, Klafter, Thomason A Chmiel Owski, Michel Nagin "Robotics Engg-an Integrated Approach" PHI 2005.
6. R.K. Mittal & I.J. Nagrath, "Robotics & Control" TMH-2007.
7. Groover. M.P. Industrial Robotics, technology, programming and application Mc-Graw Hill 2012.
8. S. K. Saha, "Introduction to Robotics", Tata McGraw-Hill Publishing Company Ltd. (2008).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of sensors and programming language in robotics applications. (Apply)
2. Analyze how robots can be used for industry and domestic applications. (Analysis)
3. To provide skills to design advanced robots to do industrial and domestic targets. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-202.1	Understand basic concepts of robotics and its peripherals.
AMTR-202.2	Apply knowledge of sensors in design of robots.
AMTR-202.3	Analyse different programming language to make robots operations more easy and advanced.
AMTR-202.4	Design advanced and humanoid robots to solve the specific needs of industry and society.
AMTR-202.5	Assess current burning research issues and future directions in the field robot based

ELECTIVE- (I-III)

AMTR-211	MECHATRONICS SYSTEMS DESIGN	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Mechatronic systems, Key elements, Mechatronic design process, Application types, Interfacing issues, Man Machine Interfaces, Safety features, optimization of Mechatronic design, Fault diagnosis.

Mathematical models, Block diagram modelling, Analogy approach, Impedance diagrams, Models for Electrical, Mechanical, Electro-mechanical and Fluid systems, System Identification, Least square method, Closed loop identification, joint input/output identification, State estimators, Model Validation.

Simulation basics, Probability concepts in simulation, Discrete event simulation, Simulation Methodology, Queuing system model components, Continuous system modelling, Monte Carlo simulation, Analysis of simulation results, Simulation life cycle.

Mass-Spring-Oscillation and Damping system, Position Control of Permanent magnet DC motor using Hall sensor and optical encoder, Auto-control system for Green House Temperature, Transducer Calibration system, Strain Gauge Weighing system, Solenoid Force-Displacement Calibration system.

Automatic Washing Machine, Hard Drive control, Auto-focusing in Digital Cameras, Active suspension in vehicles, Visual Servoing models, Thermal cycle fatigue of a Ceramic plate, pH Control system, De-icing temperature control system, Skip control of a CD player, Simulation of Rocket thrust control, Time delay Blower.

References and Text Books:

1. Devadas Shetty, Richard A.Kolkm, "Mechatronics system design, PWS publishing company, 2009.
2. Bolton, "Mechatronics – Electronic control systems in mechanical and electrical engineering, 2nd edition, Addison Wesley Longman Ltd., 2009.
3. Brian morriss, "Automated manufacturing Systems – Actuators Controls, sensors and Robotics", McGraw Hill International Edition, 2000.
4. Bradley, D. Dawson, N.C.Burd and A.J. Loader, "Mechatronics: Electronics in product and process", Chapman and Hall, London, 1999

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to construct the block diagram of any physical mechatronics device. (Apply)
2. Analyze the mathematical models for Electrical, Mechanical, Electro-mechanical and Fluid systems. (Analysis)
3. To provide the skills to develop the mathematical model of any physical model from any engineering domain and prototyping mechatronic systems using developed mathematical models.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-211.1	Understand the key elements of mechatronic systems, design optimization and identify possible applications of mechatronic systems
AMTR-211.2	Apply the analytical skills to develop mathematical model of electrical, mechanical, electro-mechanical and fluid systems.
AMTR-211.3	Analyse the developed mathematical model of electrical, mechanical, electro-mechanical and fluid systems through simulations and optimise the design
AMTR-211.4	Design the circuits to interface various sensors to mechatronic systems. Develop the control system for DC motor position control.
AMTR-211.5	Assess the current research issues and future directions in the field of automation for washing machine, digital cameras, pH controller, temperature controller etc.

AMTR-212	PC BASED AUTOMATION	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Basic Computer Architecture, Components of a PC, Serial and Parallel Communications, Parallel port, OSI Model, RS232, USB, ISA, PCI, PXI, PCI Express, GPIB Protocols. LAN, WAN and MAN Networks, RS485, RS 422, LXI Protocols, Modbus, Field bus, Ethernet, CAN bus, SCADA and DCS.

Continuous and Discrete signals, Sampling theorem, Quantization, Sampling and Hold, ADC,DAC, Resolution and Sampling Frequency, Multiplexing of input signals, Single ended and differential inputs, Sampling of Multi-channel analog signals. Concept, Timer, Counter and analog output in Universal DAQ card.

Algorithm, Flowchart, Variables & Constants, Expressions, Data types, Input output operations, Conditional Statements, Looping, Sub-programs/Functions, Arrays, Structures and Classes, Inheritance, Polymorphism, Debugging.

GUI, Graphical Programming, Data Flow techniques, Processing Data in GP, Loops and Structures, Event based & Schedule based operations, Global and Local Variables, File I/O operations, parallel processing of data, Virtual Instrument and control, VISA & SCPI.

References and Text Books:

1. Morris Mano M., Computer System Architecture, Prentice Hall of India, Third Edition, 2002.
2. John P. Hayes, Computer Architecture and Organization, McGraw Hill International, Third Edition, 1998.
3. William Stallings, “Computer Organization and Architecture”, VI Edition, Prentice Hall of India, 2003.
4. Krishna Kant, ‘Computer based Industrial Control’, Prentice Hall of India, 1997.
5. Gary Johnson, ‘LabVIEW Graphical Programming’, II Ed., McGraw Hill, 1997.
6. Sanjeev Gupta, ‘Virtual Instrumentation using Labview’ Tata McGraw Hill, 2004.
7. Jovitha Jeome ‘Virtual Instrumentation using Lab View’ PHI Learning Pvt Ltd, 2009.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to understand computer architecture, programming and debugging

- techniques. (Apply)
2. Analyze the continuous and discrete time signals. (Analysis)
 3. To learn data flow techniques and graphical programming, and to create control scheme for virtual instrumentation. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-212.1	Understand the basic computer architecture including serial and parallel communication techniques, develop the understanding of LAN, WAN and MAN Networks.
AMTR-212.2	Apply quantization and sampling techniques on discrete and continuous signals for designing various ADC and DAC.
AMTR-212.3	Analyse multi-channel analog signals and understand the concept of universal DAQ card.
AMTR-212.4	Design flowchart and program algorithms, develop the understanding of programming and debugging techniques.
AMTR-212.5	Assess current research issues and future directions in field of virtual instrumentation.

AMTR-213	SIGNAL PROCESSING IN MECHATRONICS SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Discrete- Time Signals: Sequences; representation of signals on orthogonal basis; Sampling and Reconstruction of signals

Discrete systems: Z-Transform, Analysis of LSI systems, Frequency Analysis, Inverse Systems, Discrete Fourier Transform (DFT), Fast Fourier Transform algorithm, Implementation of Discrete Time Systems.

Analog filters: Ideal filter characteristics, low pass, high pass, band pass and band stop filters, notch filters, comb filters, all-pass filters, and inverse systems. Digital filters: Design of FIR and IIR filters.

DSP Processors. Various Texas processors such as TMS320C6713, TMS320C6416, DM6437 Digital Video Development Platform with Camera, DevKit8000 OMAP3530 Evaluation Kit. Introduction to multi-rate signal processing. Application of DSP to Speech and Radar signal processing.

References and Text Books:

1. S. K. Mitra, Digital Signal Processing: A computer-Based Approach, 3/e, TMcHl, 2006.
2. A. V. Oppenheim and R. W. Shafer, Discrete-Time Signal Processing, Prentice Hall India, 2/e, 2004.
3. J. G. Proakis and D. G. Manolakis, Digital Signal Processing: Principles, Algorithms and Applications, 4/e, Pearson Education, 2007.
5. V.K. Ingle and J.G. Proakis, "Digital signal processing with MATLAB", Cengage, 2008.
6. T. Bose, Digital Signal and Image Processing, John Wiley and Sons, Inc., Singapore, 04.
7. L. R. Rabiner and B. Gold, Theory and Application of Digital Signal Processing, PH, 2005.
8. A. Antoniou, Digital Filters: Analysis, Design and Applications, Tata McH, 2003.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to understand mathematical description and representation of continuous

- and discrete time signals and systems. (Apply)
2. Analyze the discrete time systems, algorithms and analog filters. (Analysis)
 3. To provide skills to develop controller prototypes using evaluation kit of Texas processors such as TMS320C6713 and TMS320C6416. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-213.1	Understand the digital signals, their representation, sampling and reconstruction.
AMTR-213.2	Apply analytical skills to analyse discrete time systems.
AMTR-213.3	Analyse analog and digital filters such as low pass filter, high pass filter, band pass filter, FIR filter and IIR filters.
AMTR-213.4	Design controller prototypes for specific problems using evaluation kit of Texas processors such as TMS320C6713 and TMS320C6416.
AMTR-213.5	Assess current research issues and future directions in field of digital signal processing.

AMTR-214	CONTROL OF MECHATRONIC SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Modern control: Concept of states, state space model, different form, controllability, observability; pole placement by state feedback, observer design, Lunenburg observer, reduced order observer, observer based control.

Optimal control design: Solution-time criterion, control-area criterion, performance indices; zero steady state step error systems; modern control performance index: quadratic performance index, Ricatti equation.

Digital control: Sampling process, sample and hold, analog to digital converter, use of z-transform for closed loop transient response, stability analysis using bilinear transform and Jury method, digital control design using state feedback

Non-Linear Control System: Common physical non-linear system, phase plane method, system analysis by phase plane method, stability of non-linear system, stability analysis by describing function method, Liapunov's stability criterion, Popov's stability criterion.

References and Text Books:

1. K. Ogata, "Modern Control Engineering", Prentice Hall India (2002).
2. Gene F. Franklin, J. D. Powell, A E Naeini, "Feedback Control of Dynamic Systems", Pearson (2008).
3. John Van De Vegte, "Feedback Control Systems", Prentice Hall (1993).
4. Thomas Kailath, "Linear Systems", Prentice Hall (1980).
5. Alok Sinha, "Linear Systems: Optimal and Robust Control", Taylor & Francis (2007).
6. Brian D. O. Anderson and John B. Moore, "Optimal Control: Linear Quadratic Methods", Dover Publications (2007).
7. K. Ogata, "Discrete-Time Control Systems", PHI Learning (2009).
8. H.K. Khalil, "Nonlinear Systems", Prentice Hall (2001).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to develop the understanding of control system using the concept of state, controllability and observability for controlling mechatronic systems.
2. Analyze the steady-state and transient response of controllers to develop optimum control system for mechatronic systems.
3. To provide skills to design digital control system for mechatronic systems.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-214.1	Understand the concept of modern control system with the help of state space model, controllability and observability.
AMTR-214.2	Apply appropriate analytical techniques to provide optimum control solutions under steady-state and transient conditions.
AMTR-214.3	Sampling and analysis of systems using z-transform to develop the desired control system for the given control problem.
AMTR-214.4	Stability analyse of the non-linear control system using Liapunov's stability criterion.
AMTR-214.5	Assess current research issues and future directions in field of controller design for mechatronic systems .

AMTR-215	MACHINE TOOL CONTROL AND CONDITION MONITORING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Overview of automatic control in machine tools

Open loop and closed loop system in machine tools- process model formulation-transfer function control actions-block diagram representation of mechanical pneumatic and electrical systems.

Drive systems and feedback devices in machine tools

Hydraulic and Pneumatic drives, Electrical drives, A.C. Motor, D.C. Motor, Servo motor and Stepper motor. Feedback devices- Syncro, resolver, diffraction gratings, potentiometer, Inductosyn and encoders-application in machine tools.

Adaptive control and PLC: Types– ACC, ACO, Real time parameter estimation, Applications-adaptive control for turning, milling, grinding and EDM. Programmable logic controller-Functions-Applications in machine tools.

Vibration, acoustic emission / sound: Primary & Secondary signals, Online and Off-line monitoring. Fundamentals of Vibration, Sound, Acoustic Emission. Machine Tool Condition Monitoring through Vibration, Sound, Acoustic Emission, Case Studies.

Condition monitoring, through other techniques: Visual & temperature monitoring, Leakage monitoring, Lubricant monitoring, condition monitoring of Lube and Hydraulic systems, Thickness monitoring, Image processing techniques in condition monitoring.

References and Text Books:

1. Mikell P. Groover, "Automation Production system and Computer Integrated Manufacturing", Prentice Hall of India Pvt. Ltd., 2010
2. Sushil Kumar Srivastava, "Industrial Maintenance Management" S.Chand & Company Ltd., New Delhi, 1998.
3. Manfred Weck, "Hand Book of Machine Tools" – Vol.3, John Wiley & Sons, 1995.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in machine tool control and condition monitoring. (Apply)
2. Analyze how an effective control scheme should be selected for a particular process control system, and also analyze the different controller performance specifications. (Analysis)
3. Create prototype for PLC based controllers for various process control systems. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-215.1	Understand the machine tools and generalized control system. Understand the mathematical representation of real-world process control systems using block-diagrams and transfer functions.
AMTR-215.2	Apply the theoretical understanding to choose suitable motor for a particular application. Also apply suitable electric drive for different motors.
AMTR-215.3	Analyse the control systems used in process control systems. Learn programming techniques for programming logic controller (PLC).
AMTR-215.4	Design the condition monitoring systems and understand the challenges in controller design through different case studies.
AMTR-215.5	Assess current research issues and future directions in field of condition monitoring and control.

AMTR-216	MECHATRONICS IN METROLOGY	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Fundamentals concepts in metrology: Introduction to Metrology and Dimensional Metrology, Angular measurements, Form Measurement, Surface Roughness, Contour Measurements, Roundness, Tool wear measurement.

Instruments for dimensional metrology

Digital Small Tool Instruments, Caliper, Micrometer, Height Gauge, Coordinate Measuring Machines, Types, Design and applications, Opto-electronic Devices for Measurement, Vision Measuring Systems, optical comparators Applications in on-line and in-process monitoring systems, 3D Surface roughness measuring instruments, Applications of LASER in Metrology, Roundness measurement, Laser interferometer in machine tool Inspection, use of laser in on-line Inspection, Laser micrometer, Laser Alignment telescope.

References and Text Books:

1. Groover. M.P, Automation, and Computer Integrated Manufacturing, Prentice Hall Pvt Ltd 2010
2. Ulrich- Rembold, Armbruster and Ulzmann -" Interface technology for computer controlled manufacturing processes ", Marcel Dekker Pub. New York, 1993
3. N RADHAKRISHNAN, P., Computer Numerical Control (CNC) Machines, New Central Book Agency, 1992.
4. Steave Krar and Arthur Gill, CNC Technology and Programming, McGraw-Hill Publishing Company, 1990.
5. Robert. G. Seippel,-"Optoelectronics for Technology and Engineering", Prentice Hall New Jersey, 1989.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to develop conceptual understanding of metrology and instruments of dimensional metrology
2. Analyze the angular, form and contour measurements to develop small tool instruments for the mechatronic system used in metrology.
3. To provide skills to use LASER in metrology applications.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-216.1	Understand the fundamental concept of metrology and mechatronic systems used in metrology.
AMTR-216.2	Apply appropriate analytical techniques to develop measuring instruments for metrology related applications.
AMTR-216.3	Analyse the performance of various instruments used in dimensional metrology such as height gauge, coordinate measuring machines and opto-electronic devices.
AMTR-216.4	Design on-line and in-process monitoring systems for metrology applications using mechatronic systems.
AMTR-216.5	Assess current research issues and future directions in field of mechatronics used in metrology.

AMTR-217	MACHINE VISION	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction: Human vision, Machine vision and Computer vision, Benefits of machine vision. Physics of Light, Interactions of light, Refraction at a spherical surface, Thin Lens Equation.

Image acquisition: Scene constraints, Lighting parameters, Lighting sources, Selection, Lighting Techniques, Types and Selection, Machine Vision Lenses and Optical Filters, Specifications and Selection, Imaging Sensors, CCD and CMOS, Specifications, Interface Architectures, Analog and Digital Cameras, Digital Camera Interfaces, Camera Computer Interfaces, Specifications and Selection, Geometrical Image formation models, Camera Calibration.

Image processing: Machine Vision Software, Fundamentals of Digital Image, Image Acquisition Modes, Image Processing in Spatial and Frequency Domain, Point Operation, Thresholding, Grayscale Stretching, Neighborhood Operations, Image Smoothing and Sharpening, Edge Detection, Binary Morphology Color image processing.

Image analysis: Feature extraction, Region Features, Shape and Size features, Texture Analysis, Template Matching and Classification, 3D Machine Vision Techniques, Decision Making.

Machine vision applications

Machine vision applications in manufacturing, electronics, printing, pharmaceutical, textile, applications in non-visible spectrum, metrology and gauging, OCR and OCV, vision guided robotics, Field and Service Applications, Agricultural, and Bio medical field, augmented reality, surveillance, bio-metrics.

References and Text Books:

1. Emanuele Trucco, Alessandro Verri, "Introductory Techniques For 3D Computer Vision", First Edition, 2009
2. Rafael C.Gonzales, Richard.E.Woods, "Digital Image Processing Publishers", Third Edition, 2007
3. Alexander Hornberg, "Handbook of Machine Vision", First Edition, 2006
4. Eugene Hecht, A.R. Ganesan "Optics", Fourth Edition, 2001

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in machine vision applications. (Apply)
2. Analyze how images is acquainted, processed and analyzed through machine vision. (Analysis)
3. To provide skills to design machine vision based systems for manufacturing, electronics, printing, pharmaceutical and textile industries. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-217.1	Understand concepts of different varieties of visions like human, computer and machine.
AMTR-217.2	Apply image acquisition and image processing techniques.
AMTR-217.3	Analyze how to take decision based on images.
AMTR-217.4	Design advanced machine vision based systems for various industries
AMTR-217.5	Assess current major issues and future challenges in the field of machine vision.

AMTR-218	DESIGN OF MECHANISMS AND MACHINES	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Mobility analysis, Degree of Freedom (DOF), Mixed Mobility, Total, Partial and Fractional DOF, Closed and Open Chain Systems, Application of D-H representation for Kinematic parameter tables for standard robot structures.

Link coordinate diagram and arm matrix of SCARA, Alpha-II, PUMA articulated robot, standard robot, polar frame, structure robot, Enter transform solution, Arm matrix of standard Robots, Polar frame, structure robots

Structural Analysis and Synthesis of mechanisms, Alternative design solutions; Coding, evaluation and selection of optimum mechanism. Type synthesis, number synthesis and design of mechanisms. Indexes of merit; Graphical, Algebraic and Optimization techniques, Matrix methods of design and analysis; Design of function, Path and Motion Generators; Structural and Mechanical error; Design and Analysis using software like ADAMS. Manipulators- Classifications, actuation and transmission systems; Coordinate Transformation – DH notations, Inverse and Forward kinematics, Manipulator dynamics from Lagrangian and Newtonian point of view. Forces in Manipulator, manipulate Dynamics, selecting of robots for Robot Application Reliability of Robotic & Automation systems and their evaluation.

References and Text Books:

1. Andeen, G.B., “Robort Design Hand Book”, SRI International, McGraw Hill,
2. Craig, J.J., “Introduction to Robotics”, Mechanics and Control, Addison Wesley
3. Spong, M., and Vidyasagar, M. “Robot Dynamics and Control”, John Wiley, NY, 1989.
4. Venkataraman. S.T., and liberall. T., “Dextrous Robot Hands”, S
5. AppuKuttan, “Robotics”, I.K. International Publishing house

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of mechanisms and machines in robotics design. (Apply)
2. Analyze how robots can be designed and operated with coding, software and matrix methods. (Analysis)
3. To provide skills to design manipulators for the industries and research. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-218.1	Understand basic concepts of machines and mechanisms.
AMTR-218.2	Apply knowledge of mechanisms in designing of links in SCARA, PUMA, standard and structure robots.
AMTR-218.3	Analyze different alternative solutions for design of the robots.
AMTR-218.4	Design manipulators for the specific needs of the society and industries.
AMTR-218.5	Assess current issues and future directions in field of robotics.

AMTR-219	ARTIFICIAL INTELLIGENCE & EXPERT SYSTEM IN AUTOMATION	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Expert system. Architecture, knowledge base, inference engine, expert system shell, applications.
Fuzzy Logic: Fuzzy sets, membership functions, operation on fuzzy sets; fuzzy control system, Fuzzification, knowledge base, inference, defuzzification, application.

Neural Network: Neuron structure, classification, artificial neural network, back propagation training and algorithm, neuro-fuzzy controllers, applications.

Genetic algorithms: Concepts, encoding and selection methods, genetic operators (crossover and Mutation), applications.

References and Text Books:

1. Haykin “Neural Networks – A comprehensive Foundation” (Mc-millan)
2. J.M. Zureda “Introduction to artificial neural networks” (Jaico)
3. A Cichocki& R Unbehauen “Neural Networks for optimization and signal Processing” John Wiley
4. George J. Klin& Tina A Polgar “Fuzzy sets, uncertainty and Information”
5. BaertKosko “Neural network and fuzzy systems”
6. Peterson “Introduction to Artificial Intelligence and expert system (PHI)
7. Michell “Introduction to Genetic Algorithm” (PHI)
8. Vidyasagar M “Theory of learning and generalization” Springer
9. S. Rajasekaran, G.A. VijaylakshmiPai “Neural Networks, Fuzzy Logic and Genetic Algotithm”, PHI.
10. T.J. Ross: “Fuzzy Logic with Engineering Applications” Second Edition John Wiley India.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in expert systems in architecture, expert system shell and knowledge base. (Apply)
2. Analyze how fuzzy logic can be used in expert systems. (Analysis)
3. To provide skills to design neural networks based on different algorithms. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-219.1	Understand basic concepts and applications of expert systems.
AMTR-219.2	Apply fuzzy logic functions to provide artificial intelligence to any system.
AMTR-219.3	Analyze different neural networks based on different algorithms.
AMTR-219.4	Design artificial intelligence and expert systems based automated systems .
AMTR-219.5	Assess current issues and future challenges in field of artificial intelligence.

AMTR-220	MODELLING AND SIMULATION OF MECHATRONIC SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Physical Modelling: Mechanical and electrical systems, physical laws, continuity equations, compatibility equations, system engineering concept, system modelling with structured analysis, modelling paradigms for mechatronic system, block diagrams, mathematical models, systems of differential-algebraic equations, response analysis of electrical systems, thermal systems, fluid systems, mechanical rotational system, electrical-mechanical coupling.

Simulation Techniques: Solution of model equations and their interpretation, zeroth, first and second order system, solution of 2nd order electro-mechanical equation by finite element method, transfer function and frequency response, non-parametric methods, transient,

correlation, frequency, Fourier and spectra analysis, design of identification experiments, choice of model structure, scaling, numeric methods, validation, methods of lumped element simulation, modelling of sensors and actuators, hardware in the loop simulation (HIL), rapid controller prototyping, coupling of simulation tools, simulation of systems in software (MATLAB, LabVIEW) environment.

References and Text Books:

1. L. Ljung, T. Glad, "Modeling of Dynamical Systems", Prentice Hall Inc. (1994).
2. D.C. Karnopp, D.L. Margolis and R.C. Rosenberg, "System Dynamics: A Unified Approach", 2nd Edition, Wiley-Interscience (1990).
3. G. Gordon, "System Simulation", 2nd Edition, PHI Learning (2009).
4. V. Giurgiutiu and S. E. Lyshevski, "Micromechatronics, Modeling, Analysis, and Design with MATLAB", 2nd Edition, CRC Press (2009).

Course Objectives

At the end of this course, students should:

1. To introduce different types of system used in mechatronics and develop a set of algebraic equations to represent and model a complicated system into a more simplified form.
2. To employ time domain and frequency domain analysis to predict and diagnose transient and steady-state performance of the mechatronic system.
3. To analyse the transient and steady-state performance of different mechatronic systems using MATLAB and LabVIEW simulation platforms.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-220.1	Understand the need and basics of physical modelling of electrical and mechanical control systems with the help of block diagrams and mathematical model formation of a given plan.
AMTR-220.2	Apply appropriate analytical techniques to calculate the time domain specifications under steady state and transient conditions.
AMTR-220.3	Analysis of the model equations and their interpretation using MATLAB and LabVIEW.
AMTR-220.4	Understand the sensors and actuators using hardware in loop (HIL) and develop rapid prototype of mechatronic systems.
AMTR-220.5	Assess current research issues and future directions in field of mathematical and simulation based modelling of mechatronic systems.

AMTR-221	ADVANCED MICROCONTROLLER AND EMBEDDED SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	MICROCONTROLLERS AND EMBEDDED SYSTEMS				

Syllabus:

Introduction to embedded systems: Definitions – Brief overview of micro-controllers - DSPs,- Typical classifications –Memory Devices and application scenarios of embedded systems.

Arm 7 core: Introduction about ARM 7 Processor- Internal Architecture – Modes of Operations – Register set –Instruction Sets – ARM Thumb -Thumb State Registers – Pipelining – Applications.

Arm 9 core: Introduction about ARM 9 Processor-DSP Processor—Sharc Processor -- Internal Architecture – Modes of Operations – Register set – Pipelining – AMBA - Applications.

Real time models, language and operating systems: Models and languages – State Machine and state tables in embedded design – High level language descriptions - Java based embedded system design – Petrinet models-Real time languages – The real time Kernel - OS tasks - Task Scheduling - kernel services – Real time languages and their features.

Case studies in real time embedded systems: Specific examples of time-critical and safety-critical embedded systems - applications in automation -automotives – aerospace - medical and manufacturing.

References and Text Books:

1. Wayne Wolf, Computers as Components – Principles of Embedded Computing System Design, Morgan Kaufmann Publishers 2009.
2. Ball S.R., Embedded microprocessor Systems – Real World Design, Prentice Hall, 2006
3. C.M. Krishna, Kang G. Shin, Real Time systems, McGraw Hill 2009
4. Frank Vahid and Tony Givagis, Embedded System Design
5. Tim Wilmshurst, An Introduction to the design of small – scale Embedded Systems.

Course Objectives

At the end of this course, students should:

1. Apply advance techniques in embedded system design. (Apply)
2. Analyze the programming techniques for ARM 7 and ARM 9 processors. (Analysis)
3. To provide embedded system based prototypes to solve real time problems in the field of aerospace and medical manufacturing. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-221.1	Understand the digital signal processors and embedded system design.
AMTR-221.2	Apply appropriate programming techniques for microcontroller and develop the skills for interfacing different devices.
AMTR-221.3	Analyse different programming techniques for ARM 7 and ARM 9 processors.
AMTR-221.4	Design ARM 7 and ARM 9 processor based prototype systems for specific control problems.
AMTR-221.5	Assess current research issues and future directions in digital signal processor based system design.

AMTR-222	FINITE ELEMENT METHOD	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction to Finite Difference Method and Finite Element Method, Advantages and disadvantages, Mathematical formulation of FEM, Variational and Weighted residual approaches, Shape functions, Natural co-ordinate system, Element and global stiffness matrix, Boundary conditions, Errors, Convergence and patch test, Higher order elements. Application to plane stress and plane strain problems, Axi-symmetric and 3D bodies, Plate bending problems with isotropic and anisotropic materials, Structural stability, other applications e.g., Heat conduction and fluid flow problems. Idealization of stiffness of beam elements in beam-slab problems, Applications of the method to materially non-linear problems, Organization of the Finite Element programmes, Data preparation and mesh generation through computer graphics, Numerical techniques, 3D problems, FEM an essential component of CAD, Use of commercial FEM packages, Finite element solution of existing complete designs, Comparison with conventional analysis.

References and Text Books:

1. Introduction to Finite Elements in Engineering, T. R. Chandrupatla & A. D. Belegundu, 2nd Ed., PHI, 2001.
2. An Introduction to the Finite Element Method, Reddy, J. N., 2005.
3. The Finite Element Method, O. C. Zienkiewicz, 3rd Ed., McGraw-Hill, 1997.
4. The Finite Element Method in Engineering, S. S. Rao, 2nd Ed., Elmsford, Pergamon, 1989.
5. Introduction to the Finite Element Method: A Numerical Method for Engineering Analysis, Desai, C. S. and Abel, J. F., 1972
6. The Finite Element Method O.C. Zienkiewicz and R.L. Taylor McGraw Hill
7. Finite Element Procedure in Engineering Analysis K.J. Bathe McGraw Hill
8. Finite Element Analysis C.S. Krishnamoorthy Tata McGraw Hill
9. Concepts and Application of Finite Element Analysis R.D. Cook, D.S. Malsus and M.E. Plesha John Wiley
10. Finite Element and Approximation O.C. Zenkiewicz & Morgan

Course Objectives

At the end of this course, students should:

1. Learn basic principles of finite element analysis procedure.
2. Understand the theory and characteristics of finite elements that represent engineering structures.
3. Apply finite element solutions to structural, thermal, dynamic problems to develop the knowledge and skills needed to effectively evaluate finite element analyses.

Course Outcome

Upon successful completion of this course you should be able to:

AMTR-222.1	Understand the concepts behind formulation methods in FEM.
AMTR-222.2	Develop element characteristic equation and generation of global equation.
AMTR-222.3	Identify the application and characteristics of FEA elements such as bars, beams, plane and iso-parametric elements and able to apply suitable boundary conditions to a global equation for bars, trusses, beams, circular shafts, heat transfer.
AMTR-222.4	Apply boundary equations for nonlinear problems, and data preparation and mesh generation through computer graphics.
AMTR-222.5	Learn different commercial FEM packages and able to apply in solving different complex problems.

AMTR-223	MECHATRONICS IN CNC	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Computer control of machines

Classification of machine tools, types, functions and processes, Need for CNC in Machine Tools, CNC systems - configuration of the CNC system, interfacing, monitoring, diagnostics, machine data - compensations for machine accuracies Open and closed loop control Adaptive control - types, application and benefits - general configuration of adaptive control and function, reasons for process change - practical problems with adaptive control - example for feedback and adaptive control.

Mechatronics elements in CNC machine tools

Guide ways and types - mechanical transmission elements - elements for rotary motion to linear motion - ball screw and types - roller screw and types - rack and pinion - various torque transmission elements - requirements of feed drives and spindle drive Types of positional control, Control of Spindle speed, Control of slide movement and velocity. Identification and gauging of work piece. Tool locking system - ball lock mechanism and contact pressure monitoring. Automatic tool changing system - types and benefits - tool magazine.

CNC programming

M/C axis identification, Primary and Secondary, Manual Programming, Turning and Milling Programming fundamentals, fixed cycles, subroutines and other Productivity Packages- CAPP and APT in programming.

References and Text Books:

1. Groover. M.P, Automation, and Computer Integrated Manufacturing, Prentice Hall Pvt Ltd 2010
2. Ulrich- Rembold, Armbruster and Ulzmann -" Interface technology for computer controlled manufacturing processes ", Marcel Dekker Pub. New York, 1993
3. N RADHAKRISHNAN, P., Computer Numerical Control (CNC) Machines, New Central Book Agency, 1992.
4. Steave Krar and Arthur Gill, CNC Technology and Programming, McGraw-Hill Publishing Company, 1990.
5. Robert. G. Seippel, -"Optoelectronics for Technology and Engineering", Prentice Hall New Jersey, 1989.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of computer control in Machines. (Apply)
2. Analyze how machines is operated with computerized control over different machining process . (Analysis)
3. To provide skills to design programmed and controlled machines. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-223.1	Understand computer based control of machines.
AMTR-223.2	Apply mechatronics elements for various power transmission drives. .
AMTR-223.3	Analyze CNC programming required for advanced computer based control.
AMTR-223.4	Design CNC machines for various needs of the industries.
AMTR-223.5	Assess current and future changes in the field of computer controlled machines.

AMTR-224	THEORY OF MACHINING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Need and Classifications of Machining Processes; Types of Study in Machining Processes- Experimental and Theoretical; conventional machining: Mechanics of Metal Cutting: Chipping action; Cutting parameters; Orthogonal and Oblique cutting; Mechanism and Types of chips; Cutting forces and Stresses; Power and Energy; Heat and Temperature; Tool Materials and Tool Life; Tool Geometry and Mechanics of Turning, Drilling and Milling; abrasive machining: Mechanics of Metal Grinding: Grinding Wheel and Chip Formation; Grinding Forces and Power; Grinding Temperature; Advanced Abrasive Processes.

References and Text Books:

1. Materials and Processes in Manufacturing, Degarmo, J. T. Black, Prentice Hall Of India Pvt Ltd.
2. Manufacturing Processes for Engineering Materials, Kalpakjian and Schmid, Prentice Hall.
3. Principles of Machine Tools, G. C. Sen and A. Bhattacharya, New Central Book Agency
4. Theory of Metal Cutting, A. Bhattacharya.
5. Manufacturing Science : Ghosh and Mallick, East-West Press Private Limited
6. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of manufacturing in machining. (Apply)
2. Analyze how machining concepts can be used to increase better performance of the cutting tools used in industries. (Analysis)
3. To provide skills to design stable machining systems for industries. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-224.1	Understand basic concepts of machining.
AMTR-224.2	Apply machining concepts in real manufacturing systems to improve their efficiency.
AMTR-224.3	Analyze advanced machining concept based manufacturing systems.
AMTR-224.4	Design better manufacturing systems with high productivity and greater tool life.
AMTR-224.5	Assess current research issues and future challenges in the field of machining.

AMTR-225	COMPUTER AIDED MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Basic principles of automation; Extending the capabilities of conventional machines through improved devices and manipulators; Basic principles of numerical control; CNC, DNC and Machining Centers; Manual part programming (industrial controller like Fanuc/Siemens); APT programming; System devices; Adaptive control; Economics of numerical control.

Introduction to Robotics: Elements of robot anatomy; Hydraulic, pneumatic and electrical manipulators; End-effectors and their design, applications of industrial robots.

References and Text Books:

1. CAD/CAM Theory and Practice, Ibrahim Zeid, Tata McGraw Hill Publication
2. CAD/CAM Principle and Application, P. N. Rao, Tata McGrawHill Publication.
3. Users guides to Rapid Prototyping, Todd Frimm, Society of Manufacturing Engineers
4. Computer control of Manufacturing system, Yoram koren, McGraw Hill Publication.
5. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.
6. Machining and CNC Technology, M. Fitzpatrick, McGraw-Hill Publication.
7. Journey from Rapid Prototyping to Rapid Manufacturing, S Chattopadhyaya, Lambert Publication.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to understand the principles of automation in industries.
2. Analyze how to operate and control the different types of NC, CNC and DNC machine tools.
3. To provide skills to write the manual part programing and APT programing to operate the machine tools.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-225.1	Understand basic principles of automation and capabilities of conventional machines.
AMTR-225.2	Study the basic principles of automation in machine tools .
AMTR-225.3	Understand the part programing and apply the manual part programming to control the various machining processes.
AMTR-225.4	Understand the APT part programing and apply the APT part programming to control the various machining processes.
AMTR-225.5	Basics of robotics and its elements. Applications of industrial robots; Economics of robotics

AMTR-226	FLEXIBLE MANUFACTURING SYSTEM	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

FMS concept, Components of FMS, FMS Layouts, FMS planning and implementation. Tool Management Systems-Tool monitoring, Work holding devices Modular fixturing, flexible fixturing, flexibility, quantitative analysis of flexibility, application and benefits of FMS, automated material handling system, AGVs, Guidance methods, AS/RS.

Automated process planning: Group Technology, Part families, Part classification and coding, Production flow analysis, Machine cell design, Applications and Benefits of Group Technology, Structure of a Process Planning, Process Planning function, CAPP - Methods of CAPP, CAD based Process **Planning, Inventory management:** Materials requirements planning - basics of JIT.

Monitoring and quality control: Types of production monitoring system, process control & strategies, direct digital control - Supervisory computer control - computer aided quality control - objectives of CAQC, QC and CIM, contact, non-contact inspection methods, CMM and Flexible Inspection systems. Integration of CAQC with CIM.

References and Text Books:

1. Kant Vajpayee. S., "Principles of Computer Integrated Manufacturing", Prentice Hall of India, 1999.
2. Radhakrishnan.P, Subramanyan. S, "CAD/CAM/CIM", New Age International publishers, 2000.
3. Scheer. A.W., "CIM- Towards the factory of the future" Springer - Verlag, 1994.
4. Daniel Hunt.V., "Computer Integrated Manufacturing Hand Book", Chapman & Hall, 1989.
5. Groover M.P, "Computer Aided Design and Manufacturing", Prentice Hall of India, 1987.
6. Yorem Koren, "Computer Control of Manufacturing System", McGraw Hill, 1986.
7. Ranky Paul. G., "Computer Integrated Manufacturing", Prentice Hall International, 1986

Course Objectives

The main objectives of the course are:

1. To introduce and discuss flexible manufacturing concepts;
2. To have the students gain insight about the state-of-the-art research areas related to FMS and real-time shop floor control
3. To have the students gain advanced skills in modeling, design and simulation of complex systems

Course Outcome

Upon completion of their academic work student will be able:

AMTR-226.1	Apply the concepts of PPC and GT to the development of FMS.
AMTR-226.2	Identify various workstations, system support equipment's.
AMTR-226.3	Discuss the planning and scheduling methods used in manufacturing systems.
AMTR-226.4	Identify hardware and software components of FMS.
AMTR-226.5	Summarize the concepts of modern manufacturing such as JIT, supply chain management and lean manufacturing etc

AMTR-227	RAPID PROTOTYPING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction to RP processes, applications, benefits, limitations, comparison between RP processes and CNC Machining.

Different Rapid Prototyping Processes – namely Stereolithography (STL), Laser Sintering; Fused Deposition Modelling (FDM), Solid Ground Curing (SGC); Laminated Object Manufacturing (LOM) , Ballistic Particle Manufacturing (BPM), Three Dimensional Printing (3DP), numerical Examples. **Materials of RP Technology** - Photo Sensitive Resin, Wax etc.

RP interfacing with Manufacturing processes - Investment Casting, Vacuum Casting, and Laser Additive Manufacturing. Rapid manufacturing processes like - Vacuum Casting Processes via RP Processes, Electroplated Prototypes, RTV Silicon Moulds, Direct Inkjet SL Tooling, Electron Beam Melting

References and Text Books:

1. Rapid Manufacturing: An Industrial Revolution for the Digital Age. Neil Hopkinson, Richard Hague, Philip Dickens (Editors); Wiley; Jan., 2006; ISBN:10: 0470016132; 13: 978-0470016138.
2. Additive Manufacturing Technologies; Rapid Prototyping to Direct Digital Manufacturing. Ian Gibson, David W. Rosen, Brent Stucker; Springer; January, 2010; ISBN: 978-1-4419-1119-3.
3. Rapid Prototyping: Principles and Applications. Rafiq I. Noorani; Wiley; Oct., 2005; ISBN: 10: 0471730017; 13: 978-0471730019.
4. User's Guide to Rapid Prototyping. Todd Grimm; Society of Manufacturing Engineers; February, 2004; ISBN: 0-87263-697-6.
5. Rapid Prototyping - Laser-based and Other Technologies. Patri K. Venuvinod and Weiyin Ma; Kluwer Academic Publishers; October, 2003; ISBN: 1-4020-7577-4.
6. Rapid Prototyping. Andreas Gebhardt; Hanser Gardner Publications; 1st ed., June 1, 2003; (Originally published in German, 1995); ISBN: 156990281X.
7. Rapid Prototyping: Principles and Applications (2nd Edition). Chua Chee Kai, Leong Kah Fai, Lim Chu-Sing; World Scientific Pub Co; March, 2003; ISBN: 9812381171.
8. Rapid Prototyping: Theory and Practice. Ali Kamrani, Emad Abouel Nasr (Editors); Springer; 1st ed., Jan., 2006; ISBN:10: 0387232907; 13: 978-0387232904.
9. Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, D.T. Pham, S.S. Dimov

Course Objectives

At the end of this course, students should:

1. Learn the basics of additive manufacturing/rapid prototyping and its applications in various fields, reverse engineering techniques.
2. Train with different processes in rapid prototyping systems.
3. Understand about mechanical properties and geometric issues relating to specific rapid prototyping applications.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-227.1	Demonstrate the knowledge and basics of Additive Manufacturing and Rapid Prototyping technologies
AMTR-227.2	To understand the different types of error formed in STL file formation and how to overcome this error.
AMTR-227.3	Describe different RP techniques and its manufacturing defects, and control.
AMTR-227.4	To learn about the different range of materials used in additive manufacturing.
AMTR-227.5	This course makes students capable to manufacture any complex parts using additive manufacturing technique and provides understanding regarding the use of different RP interface system.

AMTR-228	MEMS AND NEMS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction to MEMS & materials, fabrication processes, MEMS sensors and actuators, scaling issues in MEMS, electro mechanics, Evolution of Microsensors & MEMS, Microsensors & MEMS applications, Microelectronic technologies for MEMS, Micromachining Technology – Surface and Bulk Micromachining, Micromachined Microsensors, Mechanical, Inertial, Biological, Chemical, Acoustic, Microsystems Technology, Integrated Smart Sensors and MEMS, Interface Electronics for MEMS, MEMS Simulators, modelling and design techniques, packaging and reliability, measurement techniques for MEMS, Conclusions & Future Trends.

Nanoelectromechanical systems (NEMS) – a journey from MEMS to NEMS, MEMS vs. NEMS, MEMS based nanotechnology – fabrication, film formation and micromachining, NEMS physics – manifestation of charge discreteness, quantum electrodynamical (QED) forces, quantum entanglement and teleportation, quantum interference, quantum resonant tunneling and quantum transport, Wave phenomena in periodic and aperiodic media – electronic and photonic band gap crystals and their applications, NEMS architecture, Surface Plasmon effects and NEMS fabrication for nanophotonics and nanoelectronics, Surface Plasmon detection – NSOM/SNOM.

References and Text Books:

1. Electromechanical Sensors and Actuators, Ilene J. Busch-Vishniac, Springer, 2008
2. Introduction to Microelectronics Fabrication, Vol. V, G. W. Neudeck and R. F. Pierret (eds.), Addison, Wesley, 1988
3. Introduction to Microelectromechanical Microwave Systems, H. J. De Loss Santos, 2nd edition, Norwood, MA: Artech, 2004
4. Microsystems Design, S. D. Senturia, Kluwer – Academic Publishers, Boston MA, 2001.
5. Principles and Applications of Nano-MEMS Physics, H. J. Delos Santos, Springer, 2008.
6. Materials and Process Integration for MEMS Microsystems, Vol. 9, Francis E. H. Tay, Springer, 2002.
7. MEMS: Introduction and Fundamentals by Mohamed Gad-el-Hak –
8. MEMS mechanical sensors by Stephen Beeby
9. Microsensors, MEMS, and smart devices by Julian W. Gardner, V. K. Varadan, Osama O. Awadelkarim
10. MEMS and microsystems: design, manufacture, and nanoscale engineering- Tai-Ran Hsu
11. MEMS by N P Mahalik.

Course Objectives

At the end of this course, students should:

1. Have a concept on the scope and recent development of the science and technology of micro- and nano-systems.
2. Gain the physical knowledge underlying the operation principles and design of micro and nano-systems
3. Learn some typical or potentially applicable micro- and nano-systems at the frontier of the development of the field

Course Outcome

Upon completion of their academic work student will be able:

AMTR-228.1	Ability to understand the operation of micro devices, micro systems and their applications
AMTR-228.2	Ability to design the micro devices, micro systems using the MEMS fabrication process.
AMTR-228.3	Gain a knowledge of basic approaches for various Nano electromechanical systems (NEMS).
AMTR-228.4	Gain a knowledge of basic approaches for quantum interference, quantum resonant tunneling and quantum transport.
AMTR-228.5	Gain the technical knowledge required for NEMS architecture, and NEMS fabrication for nano photonics and nano electronic.

AMTR-229	SMART MATERIALS FOR MECHATRONICS APPLICATIONS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction: Smart materials and their application for sensing and actuation, Mechatronics aspects.

Piezoelectric materials: Piezoelectricity and piezoelectric materials, Constitutive equations of piezoelectric materials, Piezoelectric actuator types, Control of piezoelectric actuators, Applications of piezoelectric actuators for precise positioning and scanning.

Shape memory alloys (SMA): Properties of shape memory alloys, Shape memory effects, Pseudo-elasticity in SMA, Design of shape memory actuator, selection of materials, Smart actuation and control, Applications of SMA in precision equipment for automobiles, trains and medical devices

Electro-active polymers (EAPs): Ionic polymer metal composites (IPMC), Conductive polymers, Carbon nanotubes, Dielectric elastomers, Design & control issues for EAP actuators, Applications of EAP for biomimetic, tactile display and medical devices.

Magnetostrictive materials: Basics of magnetic properties of materials, magnetostriction: constitutive equations, types of magnetostrictive materials, Design & control of magnetostrictive actuators, Applications of magnetostrictive materials for active vibration control.

Comparative analysis of different smart materials based actuators, Conclusions, Future research trend and applications trends of smart materials and smart materials based actuator technology.

References and Text Books:

1. Jose L. Pons, Emerging Actuator Technologies, a Micromechatronics Approach, John Wiley & Sons Ltd, 2005
2. Ralph Smith, Smart Material Systems: Model Development, SIAM, Society for Industrial and Applied Mathematics, 2005
3. F. Carpi, D. De Rossi, R. Kornbluh, R. Pelrine, P. Sommer-Larsen, Dielectric Elastomers as Electromechanical Transducers, Elsevier, Hungary, 2008,
4. Y. B. Cohen, Electroactive Polymer (EAP) Actuators as Artificial Muscles Reality, Potential and Challenges, SPIE press, USA, 2004.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of smart materials for their application as sensors and actuators in Mechatronics. (Apply)
2. Analyze smart materials like piezoelectric materials, shape memory alloys, electroactive polymers and magnetorestrictive materials. (Analysis)
3. To provide skills to design advanced mechatronics systems based on smart materials.. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-229.1	Understand basic concepts and the application of smart materials in Mechatronics.
AMTR-229.2	Apply knowledge of smart materials for smart applications.
AMTR-229.3	Analyze working of different smart materials material sand their application in mechatronics.
AMTR-229.4	Design smart materials as per the specific need of industries and society.
AMTR-229.5	Assess current trends, issues and future trends in the field of smart materials.

AMTR-230	OPTIMIZATION TECHNIQUES & DESIGN OF EXPERIMENTS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Single variable non-linear unconstrained optimization: One dimensional Optimization methods, Uni-modal function, elimination method, Fibonacci method, golden section method, interpolation methods- quadratic & cubic interpolation methods.

Multi variable non-linear unconstrained optimization: Direct search method, Univariant Method – pattern search methods – Powell’s – Hook – Jeeves, Rosenbrock search methods – gradient methods, gradient of function, steepest decent method, Fletcher reeve’s method, Variable metric method.

Geometric and Dynamic programming: Polynomials – arithmetic – geometric inequality – unconstrained G.P, – constrained G.P. Multistage decision process, principles of optimality, examples, conversion of final problem to an initial value problem, application of dynamic programming, production inventory. Allocation, scheduling replacement.

Linear and Integer programming: Formulation – Sensitivity analysis. Change in the constraints, cost coefficients, coefficients of the constraints, addition and deletion of variable, constraints. Gomory cutting plane algorithm, Zero or one algorithm, branch and bound method.

Stochastic programming: Basic concepts of probability theory, random variables – distributions – mean, variance, Correlation, co variance, joint probability distribution – stochastic linear, dynamic programming.

References and Text Books:

1. Optimization theory & Applications/ S.S Rao/ New Age International
2. Introductory to operation research/Kasan& Kumar/Springer
3. Optimization Techniques theory and practice / M.C Joshi, K.M Moudgalya/ Narosa Publications.
4. Operation Research/H.A. Taha/TMH
5. Optimization in operations research/R.L Rardin.
6. Optimization Techniques/Benugundu & Chandraputla/Person Asia.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in optimization techniques and design of experiments. (Apply)
2. Analyze how single variable and multi-variable optimization methods can be used as an effective tool for real time problems. (Analysis)
3. To provide skills to design security protocols for recognize security problems. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-230.1	Understand the concept of state variable and non-linear unconstrained optimization methods.
AMTR-230.2	Apply appropriate skills to understand multi-variable optimization methods for solving real time problems.
AMTR-230.3	Analyse the geometric and dynamic programming techniques and explore the possible applications of dynamic programming.
AMTR-230.4	Develop the understanding of sensitivity analysis and simulation of linear programming techniques.
AMTR-230.5	Assess current research issues and future directions in field of stochastic programming.

AMTR-231	ADDITIVE MANUFACTURING & TOOLING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction: Historical developments, Fundamentals of RP Systems and its Classification, Rapid prototyping process chains, 3D modeling and mesh generation, Data conversion and transmission.

RP Systems: Liquid polymer based rapid prototyping systems, Teijin Seikis' solid form and other similar commercial RP systems, Solid input materials based rapid prototyping systems, laminated object manufacturing (LOM) and fused deposition modelling systems etc., Power based rapid prototyping systems, selective Laser sintering, Soligen Diren's shell production casting (DSPC), Fraunhofer's multiphase jet solidification (MJS) and MIT's 3D printing (3DP) etc.

RP Database: Rapid prototyping data formats, STL format, file problems and file repair. Network based operations, Digital inspection. Data warehousing and learning from process data. **RP**

Applications: Development of dies for moulding, RP applications in developing prototypes of products, application in medical fields, Development of bone replacements and tissues, etc., RP materials and their biological acceptability.

References and Text Books:

1. Rapid Prototyping of Digital Systems: A Tutorial Approach, Hamblen James O, Kluwer Aca
2. Rapid Prototyping: Principles and Applications, Kai Chua Chee, World Science
3. Rapid System Prototyping with Fpgas: Accelerating the Design Process, R C Cofer, Newnes
4. Rapid Prototyping of Digital Systems, James O Hamblen, Springer.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of manufacturing, 3D modeling and design in rapid prototyping.(Apply)
2. Analyze how rapid prototyping can be used as an effective tool in increasing effectiveness and efficiency of the existing manufacturing systems. (Analysis)
3. To provide skills to design advanced manufacturing system using rapid prototyping concepts.. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-231.1	Understand basic concepts of rapid prototyping and 3D printing.
AMTR-231.2	Apply appropriate design techniques to speed up rapid prototyping with greater efficiency.
AMTR-231.3	Analyse different methods to prepare a robust database for rapid prototyping.
AMTR-231.4	Design rapid prototyping based advanced systems for medical, manufacturing and polymer industries.
AMTR-231.5	Assess current challenges and future potential in the field of rapid prototyping.

AMTR-232	VETRONICS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Fundamentals of vehicle engineering

Engine, Types, Modern Engines, Advanced GDI, Turbo-charged engines Transmissions, Chassis systems, Need for Avionics in Civil and Military aircraft and Space systems **Automotive engine control, monitoring and diagnostics systems**

Components of Electronic Engine Management, Engine control functions, Engine control modes, Fuel delivery systems, MPFI, Ignition Systems, Diagnostics, Compression Ignition Engines, Emission control Management, Hybrid Power Plants, BAS.

Automotive transmission and safety systems

Transmission control, Autonomous cruise control, Braking control, ABS, Traction control, ESP, ASR, Suspension control, Steering control, Stability control, Parking Assist Systems, Safety Systems, SRS, Blind Spot Avoidance, Auto transmission electronic control, Telematics, Automatic Navigation, Future Challenges.

Aircraft mechatronics

Fundamentals - components of an airplane and their functions - motions of a plane, Inertial Navigation, Sensors - Gyroscope- Principles. Types of actuation Systems-Linear and non-linear actuation system. Satellite navigation - GPS -system description - basic principles -position and velocity determination

Marine mechatronic systems

Basics of Marine Engineering – Marine Propulsion Mechatronics elements in ships, submarines, Variable Buoyancy Systems.

References and Text Books:

1. William B. Ribbens, "Understanding Automotive Electronics – 7th Edition, Butterworth, Heinemann Woburn, 2004.
2. Robert N Brady, Automotive Computers and Digital Instrumentation, Areston Book Prentice Hall, Eagle Wood Cliffs, New Jersey, 2000..
3. R.K. Jurgen, Automotive Electronics Handbook, McGraw Hill 2nd Edition. 2000
4. Collinson R.P.G. 'Introduction to Avionics', Chapman and Hall, 2002
5. Pallet, E.H.J. 'Aircraft Instruments & Integrated systems', McGraw-Hill, 2002
6. Myron Kyton, Walfred Fried, 'Avionics Navigation Systems', John Wiley & Sons, 2000.
7. Pallett, E.H.J. 'Aircraft instruments, principles and applications', Pitman publishing Ltd., London, 1995

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in learning fundamentals of vehicle engineering. (Apply)
2. Analyze how vehicle engineering can be used for aircraft and marine mechatronics systems. . (Analysis)
3. To provide skills to design automotive engine control, automotive transmission and safety systems. . (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-232.1	Understand fundamentals of vehicles engineering.
AMTR-232.2	Apply concepts of vehicle engineering in automotive engine control, automotive transmission, diagnostic and safety systems.
AMTR-232.3	Analyze aircraft and marine mechatronics systems. .
AMTR-232.4	Design advanced automotive systems for military and civil applications.
AMTR-232.5	Assess current issues and future directions in the field of vehicle engineering.

MMTR-233	COMPUTATIONAL FLUID DYNAMICS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Concept of Computational Fluid Dynamics: Different techniques of solving fluid dynamics problems, their merits and demerits, governing equations of fluid dynamics and boundary conditions, classification of partial differential equations and their physical behavior, Navier-Stokes equations for Newtonian fluid flow, computational fluid dynamics (CFD) techniques, different steps in CFD techniques, criteria and essentialities of good CFD techniques.

Finite Difference Method (FDM): Application of FDM to model problems, steady and unsteady problems, implicit and explicit approaches, errors and stability analysis, direct and iterative solvers. Finite Volume

Finite Volume Method (FVM): FVM for diffusion, convection-diffusion problem, different discretization schemes, FVM for unsteady problems.

Prediction of Viscous Flows: Pressure Poisson and pressure correction methods for solving Navier-Stokes equation, SIMPLE family FVM for solving Navier-Stokes equation, modelling turbulence.

CFD for Complex Geometry: Structured and unstructured, uniform and non-uniform grids, different techniques of grid generations, curvilinear grid and transformed equations. Lattice Boltzman

References and Text Books:

1. J. D. Anderson, "Computational Fluid Dynamics", McGraw-Hill Inc. (1995).
2. S. V. Patankar, "Numerical Heat Transfer and Fluid Flow", Hemisphere Pub. (1980).
3. K. Muralidhar, and T. Sundarajan, "Computational Fluid Flow and Heat Transfer", Narosa (2003).
4. D. A. Anderson, J. C. Tannehill and R. H. Pletcher, "Computational Fluid Mechanics and Heat Transfer", Hemisphere Pub. (1984).
5. M. Peric and J. H. Ferziger, "Computational Methods for Fluid Dynamics", Springer (2001).
6. H. K. Versteeg and W. Malalaskera, "An Introduction to Computational Fluid Dynamics", Dorling Kindersley (India) Pvt. Ltd. (2008).
7. C. Hirsch, "Numerical Computation of Internal and External Flows", Butterworth-Heinemann, (2007).
8. J. M. Jaile, "Molecular Dynamics Simulation: Elementary Methods", Willey Professional, 1997.
9. A. A. Mohamad, "Lattice Boltzman Method: Fundamentals and Engineering Applications with Computer Codes", Springer (2011).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge to understand the different computational fluid dynamics techniques to solve the fluid dynamics problems.
2. Analyze and develop the models to for steady and unsteady problems, implicit and explicit approaches and convection-diffusion to solve the problems.
3. To provide skills to model and simulate the any fluid mechanics related problem for complex geometries.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-233.1	Basic understanding of computational fluid dynamics its fundamental governing equations and different solving techniques.
AMTR-233.2	Different approaches to solve the problems such as finite difference method and finite volume methods.
AMTR-233.3	Prediction and calculation of different constants and parameters for solving navier-stokes equation.
AMTR-233.4	Apply the computational fluid dynamics approach on complex geometries under conditions.
AMTR-233.5	Basics of molecular dynamics algorithms and calculation of potential and force using boltzman equation, lattice boltzman equation.

MMTR-234	INTELLIGENT VISUAL SURVEILLANCE SYSTEMS	L 3	T 0	P 0	C 3
Prerequisite	DIGITAL SIGNAL PROCESSING				

Syllabus:

Basics of Image Processing: Introduction to Image Processing methods, Image Transforms, Wavelet Transform, JPEG Image Compression, Image Formats, Color Spaces- RGB, CMY, HSI.

Video Compression Standards: H. 261, H. 263, H.264, MPEG-1, MPEG-2, MPEG-4, MPEG-7, and MPEG-21, Video shot boundary detection, motion modeling and segmentation techniques.

Object Detection and Classification- Shape based object classification, motion based object classification, Silhouette-Based Method for Object Classification, Viola Jones object detection framework, Multiclass classifier boosting.

Multi-Object Tracking- Classification of multiple interacting objects from video, Region-based Tracking, Contour-based Tracking, Feature-based Tracking, Model-based Tracking, Hybrid Tracking, Particle filter based object tracking, Mean Shift based tracking, Tracking of multiple interacting objects.

Human Activity Recognition- Template based activity recognition, Sequential recognition approaches using state models (Hidden Markov Models), Human Recognition Using Gait, HMM Framework for Gait Recognition, Description based approaches, Human interactions, group activities, Applications and challenges.

Camera Network Calibration - Types of CCTV (closed circuit television) camera- PTZ (pan-tilt zoom) camera, IR (Infrared) camera, IP (Internet Protocol) camera, wireless security camera, Multiple view geometry, camera network calibration, PTZ camera calibration, cameraplacement, smart imagers and smart cameras.

References and Text Books:

1. Murat A. Tekalp, "Digital Video Processing", Prentice Hall, 1995.
2. Y. Ma and G. Qian (Ed.), "Intelligent Video Surveillance: Systems and Technology", CRC Press, 2009.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge in image processing and visual surveillance systems.
2. Analyze how compression techniques can be used for image and video compression.
3. To provide skills to design tracking and recognition systems.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-234.1	Understand the basic digital signal processing techniques with the help of wavelet transform.
AMTR-234.2	Apply appropriate skills to understand how compression techniques can be used for image and video compression.
AMTR-234.3	Analyse different multi-object tracking systems such as Region-based Tracking, Contour-based Tracking, Feature-based Tracking, Model-based Tracking, Hybrid Tracking, and Particle filter based object tracking.
AMTR-234.4	Design the human activity recognition systems and investigate the possible applications and challenges of human activity recognition systems.
AMTR-234.5	Assess current research issues and future directions in field of camera network calibration.

AMTR-235	BIO-MECHATRONICS	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Bio Mechanics Cardiovascular biomechanics, Musculoskeletal and orthopedic biomechanics, human ergonomic, Rehabilitation. Bio Sensors and Actuators.

Introduction to Bio mechatronics, Electrodes - Types, - Measurement of blood pressure - Blood Gas analyzers: pH of blood, Smart actuators for biological applications.

Medical Measurements Heart rate - Heart sound -Pulmonary function measurements -spirometer -finger-tip oximeter - ESR, GSR measurements.

Signal Processing Bio-medical signals, Signal acquisition and signal processing-Isolation barriers, Bio-Image processing.

Sensory Assist Devices Hearing aids – Implants, Optical Prosthetics, Visual Neuroprostheses – Sonar based systems, Respiratory aids, tactile devices for visually challenged.

Active and Passive Prosthetic Limbs Introduction to prosthetics, Passive Prosthetics – walking dynamics, Knee and foot prosthesis. Active prosthesis - Control of Prosthetic Arms and Hands, Leg Mechanisms, Ankle–Foot Mechanisms, Prosthesis Suspension

Wearable mechatronics devices Wearable Artificial Kidney, Wireless capsule endoscope, Wearable Exoskeletal rehabilitation system, Wearable hand rehabilitation,

References and Text Books:

1. Graham M. Brooker, “Introduction to Bio-Mechatronics”, Sci Tech Publishing, 2012.
2. Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, “Bio-Medical Instrumentation and Measurements”, II edition, Pearson Education, 2009.
3. Raymond Tong Kaiyu . “Bio-mechatronics in Medicine and Healthcare” Pan Stanford Publishing, CRC Press, 2011.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of bio mechanics in bio sensors and actuators. (Apply)
2. Analyze how bio medical signals can be acquainted and processed. (Analysis)
3. To provide skills to design wearable artificial kidney, wearable exoskeletal rehabilitation systems and wireless capsule endoscope using knowledge of mechatronics. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-235.1	Understand bio mechanics basic principles and its application.
AMTR-235.2	Apply bio mechanics concepts in making of various medical devices.
AMTR-235.3	Analyze acquisition and processing of medical signals in bio mechatronics.
AMTR-235.4	Design active and passive prosthetic limbs and their control mechanisms. .
AMTR-235.5	Assess current issues and future directions in field of bio mechatronics.

AMTR-236	INTERNET OF THINGS AND SMART MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Concept of Internet of Things (IoT), common definitions, applications, and functional view. Internet Technology Cloud Computing, Semantic Technologies, Networking and Communication Technologies.

Contemporary Manufacturing Paradigms- Architecture of IoT-MS, Integration framework of Real-time manufacturing information, Work logic of IoT-MS.

Cloud based Manufacturing Resource configuration- Cloud manufacturing, Real-time production information perception and capturing, Cloud service selection, Cloud Machine model.

Smart Factory and Smart Manufacturing- Industry 4.0 standard, Real-time information based scheduling, capacity planning, material planning, Real-time production monitoring techniques with smart sensors, Configuration of smart shop floor, traceability and call back of defective products

References and Text Books:

1. Yingfeng Zhang, Fei Tao, Optimization of Manufacturing Systems using the Internet of Things, Academic Press- Technology & Engineering, 2016.
2. Jiafu Wan, Iztok Humar, Daqiang Zhang, Industrial IoT Technologies and Applications, Springer, 17-Aug-2016.
3. K. Wang, Y. Wang, J.O. Strandhagen, T. Yu, Advanced Manufacturing and Automation V, WIT Press, 2016.
4. Ovidiu Vermesan and Peter Friess, Internet of Things – From Research and Innovation to Market Deployment, River Publishers, 2014.

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge of manufacturing in internet of things (IOT). (Apply)
2. Analyze how IOT can be linked with conventional manufacturing systems to increase their productivity and efficiency. (Analysis)
3. To provide skills to design IOT based manufacturing system for industries. (Evaluate, Create)

Course Outcome

Upon completion of their academic work student will be able:

AMTR-236.1	Understand basic concepts of IOT and related mechanisms.
AMTR-236.2	Apply IOT knowledge in manufacturing systems to improve their functioning.
AMTR-236.3	Analyze IOT based and cloud based manufacturing systems.
AMTR-236.4	Design smart factories using smart manufacturing.
AMTR-236.5	Assess current research issues and future directions in field of IOT based manufacturing system.

AMTR-237	AUGMENTED REALITY	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Introduction to Augmented Reality (AR): History, fundamental concept and components, primary features and present development, key technology, System structure.

3D Computer Graphics & Virtual Environment for AR: Basics of 3D computer graphics; creating 3D objects and a virtual environment for AR. Components, types, platforms, and devices of AR systems.

Interactions in AR: Principal components of interaction in AR systems, types and testing.

Application: Collaborative Augmented Reality, AR and VR together; Future of interactions in AR and AI; location-based experiences, Mobile Augmented Reality. Health issues in using AR.

References and Text Books:

1. Doug A. B., Kruijff E., LaViola J. J. and Poupyrev I., 3D User Interfaces: Theory and Practice , Addison-Wesley (2005,2011p) 2nd ed.
2. Schmalstieg D. and Hollerer T., Augmented And Virtual Reality, Addison-Wesley (2016).
3. Aukstakalnis S., Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR, Addison-Wesley (2016).

Course Outcome

Upon completion of their academic work student will be able:

AMTR-237.1	Understand basic concepts of Augmented reality.
AMTR-237.2	Apply augmented reality concepts in collaborative location based experiences.
AMTR-237.3	Analyze virtual environments requirements for Augmented reality.
AMTR-237.4	Design smart and mobile devices based on Augmented reality concepts.
AMTR-237.5	Assess current challenges and future directions in field of Augmented reality intergartion with Artificial intelligence.

AMTR-238	DRONE TECHNOLOGY	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Drones: Unmanned Aircraft Systems, History, Classification, System Composition, Aerodynamics. Fundamentals of flight, DGCA safety regulations.

Drone's mechanical parts: Design and Selection of the System, Identification and selection of different components, parts, block of the drone and its function & their interconnectivity, Recognize basic principles of flying like Bernoulli's Principle. Types of propeller designs used in drones.

Drone's electronic parts: Different electronic components and their measurement, test different sensors, Identify, test and install different sensors on drone. Type of batteries, specifications and their charging techniques. DC, BLDC and servo motors and test driving circuits. Drone Electronic Speed Controller (ESC).

Communication, Payloads and Controls: Inspect, test and execute GPS navigation and telemetry module, different RF blocks and antennas used in RF transmitter and receiver. Payloads, controls-PID feedback, radio control frequency range, modems, memory system, simulation, ground test-analysis-trouble shooting.

Navigation, Testing and Drone pilot training: Waypoints navigation, ground control software, System Ground Testing, System In-flight Testing, Calibrate and troubleshoot drone gimbal and drone payload, Identify and resolve common error messages and corrections by software debugging, Drone equipment fabrication, Future Prospects and Challenges.

References and Text Books:

1. Reg Austin "Unmanned Aircraft Systems UAV design, development and deployment", Wiley, 2010.
2. Robert C. Nelson, Flight Stability and Automatic Control, McGraw-Hill, Inc, 1998.
3. Kimon P. Valavanis, "Advances in Unmanned Aerial Vehicles: State of the Art and the Road to Autonomy", Springer, 2007
4. Paul G Fahlstrom, Thomas J Gleason, "Introduction to UAV Systems", UAV Systems, Inc, 1998
5. Dr. Armand J. Chaput, "Design of Unmanned Air Vehicle Systems", Lockheed Martin Aeronautics.
5. https://dgt.gov.in/sites/default/files/DRONE%20_TECHNICIAN%20_NSQF%20_LEVEL_4.pdf

Course Outcome

Upon completion of their academic work student will be able:

AMTR-238.1	Understand basic concepts of Drones and its types.
AMTR-238.2	Apply navigation concepts in drones pilot training.
AMTR-238.3	Analyze communication, pay loads and controls in drones.
AMTR-238.4	Design and selection of electronic and mechanical parts of drones.
AMTR-238.5	Assess future prospects and challenges in field of drone fabrication.

AMTR-251	Robotics and Automation Lab	L 0	T 0	P 3	C 1
Prerequisite	NIL				

List of Experiments

1. Introduction to 6-axis KUKA robots.
2. Jogging of the 6-axis KUKA robots.
3. Manual Motion of End effector for point P1 to point P2 by Smartpad (Jogging Pad)
4. Pre-program motion of end effector from point T1 to T2.
5. Program motion of end effector from point T1 to T2 in looping manner.
6. Study of the operation of an ASRS.
7. Study the operation of Lathe machine.
8. Study the operation of AGV

Course Objective

The main objectives of course are:

1. Lab allows students to design, test and build autonomous and automated systems that are used in modern manufacturing in industries.
2. Teach the fundamentals of manipulator kinematics, dynamics, control and programming with a close link to the real robots used in industry.
3. Practical implementation of the theoretical and simulation based understanding of the robotic systems.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-251.1	The students can interact directly with devices such as industry standard sensors and actuators through Human Machine Interfaces (HMIs) and PCs. Gather, process and monitor experimental data locally or at remote stations/locations.
AMTR -251.2	Student can learn advanced controllers and their applications in industry.
AMTR -251.3	Understand the relevance of teaching material to the real world systems that are currently used in industry, which will ultimately enhance their employability.
AMTR -251.4	Understand the basic process for programming the industrial manipulator. This is based on understanding the different links, frames or coordinate systems of the manipulator.
AMTR -251.5	Explain Cartesian controllers of the manipulator, using the forward and inverse kinematics and dynamics of the manipulator. Understand joint controller in the Robot Operating System. Setting each joint of the arm to the values specified by the inverse kinematic model, and allowing them to specify arbitrary positions and orientation in the world, and driving the manipulator to specified position.

AMTR-252	Independent Study & Research-II (ISR-II)	L 0	T 0	P 3	C 1
Prerequisite	NIL				

List of Activities:

In this course, student will do a variety of activities.

1. The student will work on the problem identified in ISR1 and to prepare a manuscript based on the works done.
2. The students are encouraged for publication in SCI/Scopus indexed journals and conferences.
3. This course will provide enough wisdom and idea to the students for starting and selecting a good problem for the thesis dissertation.
4. Progress will be checked once a month for support and direction. A mentor will be assigned to each student for supervision.

Course Objective

The main objectives of course are:

1. The student will learn to work on research problems.
2. The basic purpose of this lab is to prepare students for good research works.
3. One of the objectives is also to help students in selection of their thesis dissertation topic.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-253.1	Understand basic concepts of research.
AMTR -253.2	Apply previous literature review in selection of a workable research problem.
AMTR -253.3	Analyze the methods through which a research problem can be solved.
AMTR -253.4	Design a scheme to do work on research problem and to achieve targets.
AMTR -253.5	Assess challenges in doing research work based on literature reviews.

SEMESTER-III
CORE COURSES

AMTA-301	Research Methodology and Technical Writing,	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Research Methodology:

Research process, types of research, problem identification and hypothesis formulation, Research design, methods of data collection, data analysis, reliability and validity, data presentation, and report preparation.

Introduction to Research Communication:

Grammar and Rhetoric: Sentential and supra sentential structure, Narrative and structuring argument, common error in composition. Reading skills for literature review: Previewing techniques, understanding the gist of an argument, identifying the topic sentence.

Writing skills (Part-I)

Sentence formation, Use of appropriate diction, paragraph and essay writing, coherence and cohesion. Summarizing, paraphrasing, outlining, Non-linear description, Narrative, Instruction and reporting. Descriptive and explanatory, analytical and argumentative writing, enhancing editing skills, punctuation.

Writing skills (Part-II)

Introduction to terminology, concept of research. Preparing research proposal/ Synopsis. Formulating thesis statement. Referencing (all style sheet). Writing Introduction, Footnotes/ Endnotes, Conclusion. Preparing Appendix, Bibliography (all style sheets), and Abstract. Writing acknowledgement. Concept of Keywords, preparing content page/ list of Tables and Figures. Use of classified materials, Plagiarism and copyright materials.

References and Text Books:

1. Ranjit Kumar, research methodology: a step-by-step guide for beginners, SAGE Publications India Pvt Ltd.
2. C.R. Kothari, Research Methodology: Methods and Techniques, New Age International, 2004.
3. Michael H. Markel. Handbook of technical writing. Bedford/St Martins, 2012
4. Technical Writing and Professional Communication: For Nonnative Speakers of English. Thomas N. Huckin. McGraw Hill 1991.
5. Michael H. Markel. Technical Writing Essentials. St. Martin's Press, 01-May-1988.
6. Sharon J Gerson, Steven M Gerson. Technical writing: Process and product. Fifth edition, Pearson Education.

Course Objectives

At the end of this course, students should:

1. Develop skills in the use of tools and techniques required for conducting qualitative research.
2. Plan, conduct, present and discuss qualitative research study.
3. Apply the skills to writing proposals and research/project reports.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-301.1	Independently design and conduct a qualitative research regarding methods of data collection, data presentation, and report preparation.
AMTR-301.2	Describe, and apply qualitative research tools like research communications to explore the reading skills for literature review interviewing.
AMTR-301.3	Understand the skills of sentence formation, use of appropriate diction, paragraph, summarizing, paraphrasing, and outlining,
AMTR-301.4	Write preparing research proposal/ synopsis, formulating thesis statement and referencing (all style sheet).

AMTR-351	Thesis/Dissertation-I	L 0	T 0	P 18	C 9
Prerequisite	None				

Course Contents

The student will be encouraged to finalize the area of the project work during the end of second semester. The project works related to industry specific social relevant problem solving will be encouraged. The project can be carried out in the University or in collaboration with an industry/research organization/other University of repute. If a student undergoes his/her project work outside the institute, then one Supervisor will be from the collaborative institute/industry/ research organization and one faculty member from the Centre will act as an Internal Supervisor with the approval of the competent authority of the institute. At the end of the third semester, students will have to present a seminar on the progress of his/her research work. A brief project report needs to be submitted. There shall be evaluation of the work carried out at the end of the third semester.

Course Objectives

At the end of this course, students should:

1. To provide students with opportunity to learn new concepts and skills acquired in core courses and further extend these ideas to solve research/industry related problems.
2. The students will identify a research area of interest and a relevant research problem to work on.
3. A list of objectives and deliverables along with the timeline are also identified.
4. Research Progress Seminar will be held twice a month for continuous evaluation.
5. The students will continue to work on the problem identified in Major Project (in Semester IV) also.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-351.1	Explain factual knowledge (terminology, classifications, methods, trends) of current areas of research.
AMTR-351.2	Understand the steps for systematic research and the parameters involved in the research process.
AMTR-351.3	State and explain some fundamental principles, generalizations, or theories to solve the research problem.
AMTR-351.4	Complete implementation of an appropriate research design by achieving the specific skills and competencies to draw appropriate conclusions from research work.
AMTR-351.5	To evaluate the significant contribution and identify promising new directions of the work.

SEMESTER- IV

AMTR-451	Thesis/Dissertation-II	L 0	T 0	P 26	C 13
Prerequisite	None				

Course Code: AMTR-451

Course Name: Thesis/Dissertation-II,

Credit: 13

Syllabus

The students may continue to work on the problem identified in “Thesis/Dissertation- I” (in Semester III) as per the work plan. The work is continued until all stated objectives and deliverables are met. Student will prepare a comprehensive report containing introduction to the problem, literature review, methodology, results and discussion and conclusion.

Research Progress Seminar will be held periodically during the semester as per the instructions from the competent authority of Program/institution for continuous evaluation and at the end of the semester, a final viva-voce exam will be conducted for evaluating the thesis.

Course Objectives

At the end of this course, students should:

6. Design of the formulated solution and develop skills in the use of tools and techniques required for conducting qualitative research.
7. Plan and conduct experimental procedures, test procedures, data collection.
8. Analysis of the results and apply the skills to writing proposals and research/project reports.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-451.1	Formulate a research problem relevant to society/industry through critical review of appropriate and relevant information sources.
AMTR-451.2	Systematically identify relevant theory towards a research related problem, relate these to appropriate methodologies and evidence and perform experiments/modelling, etc.
AMTR-451.3	Apply qualitative and/or quantitative evaluation method to obtain the outcome of the research work undertaken.
AMTR-451.4	Complete implementation of an appropriate research design and associated methods rigorously and draw appropriate conclusions from research work.
AMTR-451.5	Prepare summary of research work in a coherent and convincing way by providing concrete findings and key arguments in the form of thesis report. Submit the report regarding conference participation or Journal publication with experimental result, result analysis.